

5G New Radio (5G NR)

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Link

http://bit.ly/5GLingga



About Me

Lingga Wardhana, ST., MBA

Educational Background :
- Electrical Engineering, Gadjah Mada University (2001 – 2006)
- MBA Program, Gadjah Mada University (2011 – 2014)

Professional Experience :
- PT. Siemens Indonesia, Network Optimization Engineer (2006 – 2008)
- PT. Lexcorp Solution, RF Optimization Consultant (2008 – 2009)
- PT. Nexwave, RF Optimization Consultant (2009)
- PT. Lintas Media Telekomunikasi, 3G RF Senior Optimization Consultant (2009 – 2010)
- PT. Floatway Systems, Founder (2010 – Present)

Achievements :
- 1st winner of Indosat Telco Project 2005
- 1st winner of Indosat Wireless Innovation Contest 2007, Hardware Category
- 3rd position in European Satellite Navigation Competititon 2009 for Regional Gipuzkoa/Spain (www.galileo-masters.eu)

Books :
- Belajar Sendiri Mikrokontroler AVR ATMega8535 (2006)
- Technopreneur (2010)
- Teknologi Wireless Communication dan Wireless Broadband (2010)
- 2G/3G RF Planning and Optimization for Consultant (2011)
- 4G Handbook Edisi Bahasa Indonesia (2014)
- Enterpreneurial MBA (2014)



Profil Learning Center

PT FLOATWAY SYSTEMS

PT. Floatway Systems (Floatway) is a company that specializes in Digital Technology and Telecommunications Training Services. Since its establishment in 2010 of over 2000 graduates have followed the Training Program

- PT. Floatway Systems (Floatway) adalah perusahaan yang didirikan dengan semangat untuk memajukan inovasi dan riset di bidang teknologi teknik elektro di negara Indonesia.
- Floatway berusaha menjadi jembatan yang sangat kokoh untuk menghubungkan problem – problem yang terjadi di lingkungan industri dan masyarakat umum dengan solusi – solusi kreatif yang dihasilkan oleh universitas dan dunia pendidikan.
- Floatway berniat menjalin hubungan baik dengan pelaku industri juga masyarakat dan terus meningkatkan pelayanan, inovasi dan mutu produk
- Di sisi pengembangan dan riset Floatway terus memberikan trigger kepada pelaku dunia pendidikan agar lebih terpacu untuk menjadi pemecah masalah – masalah real yang terjai di lingkungan bisnis dan masyarakat. Divisi learning Center akan berfungsi sebagai pembuka wawasan dan pendobrak pintu penghalang antara dunia pendidikan dan dunia aplikasi nyata

- 2G & 3G Network Capacity Optimization
- MapBasic Application
- Training 2G + 3G + 4G
- Transmission Design with Pathloss

more info: http://floatway.com/

Innovation and research
are our core values

Number of Trainee (In person)	2010	2011	2012	2013	2014
	5	147	143	171	218

Training Times	Transmission Design	VBA and Mapbasic Training	Drivetest Training	2G,3G,4G RF Planning and Optimization
	1	7	13	48



OUR CUSTOMER



























PT. MAC SARANA DJAYA









UDAYANA



AKATEL

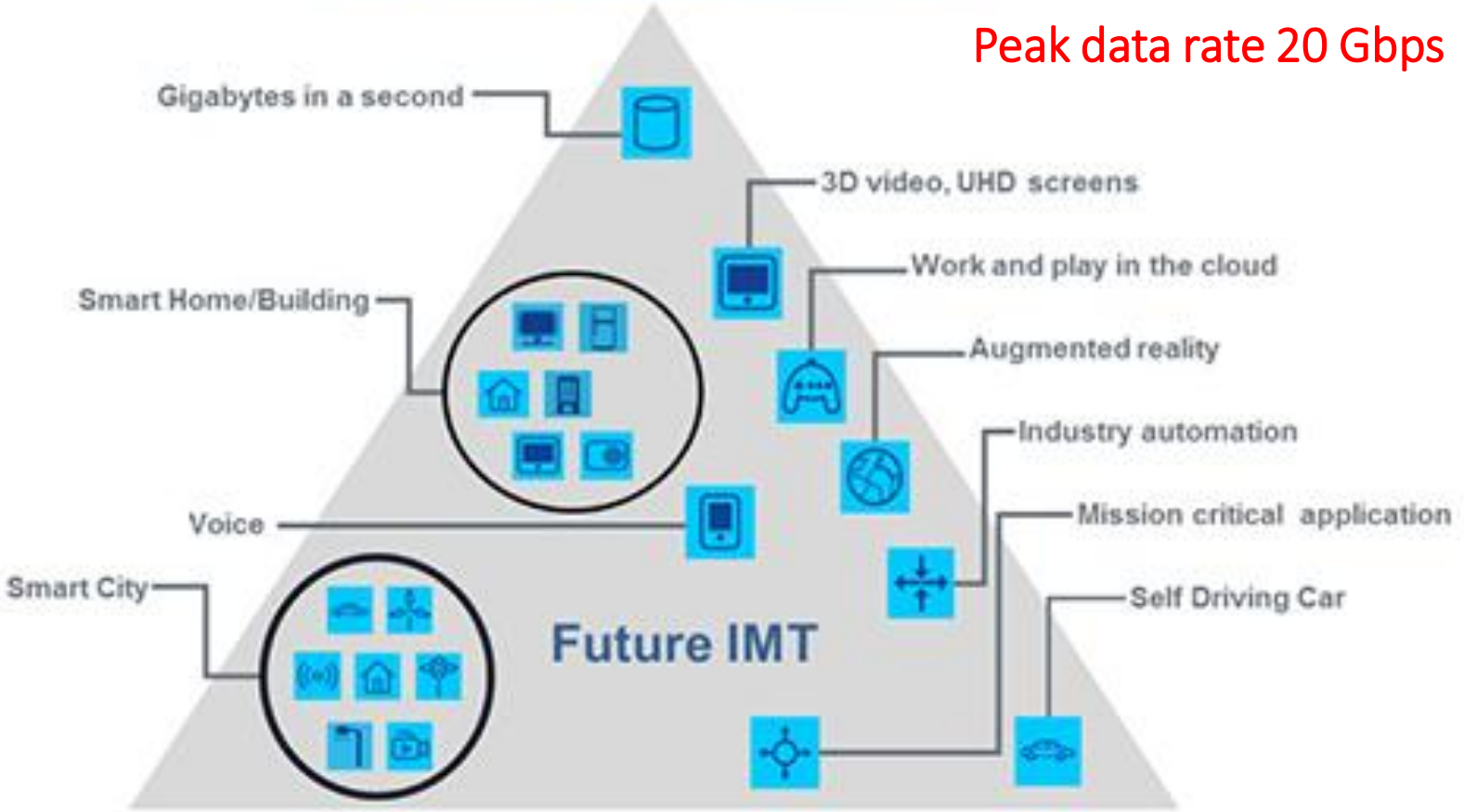


What is 5G



Enhanced Mobile Broadband (eMBB)

Peak data rate 20 Gbps



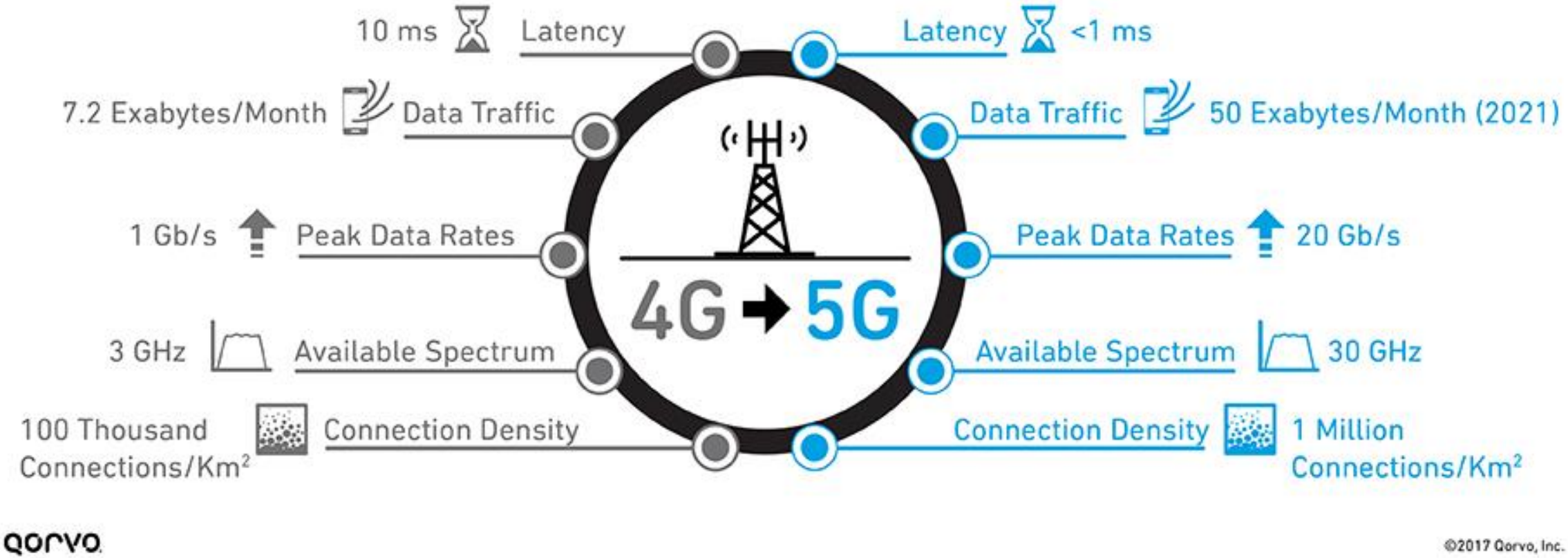
Massive Machine Type Communications (mMTC)

1 million device per km^2

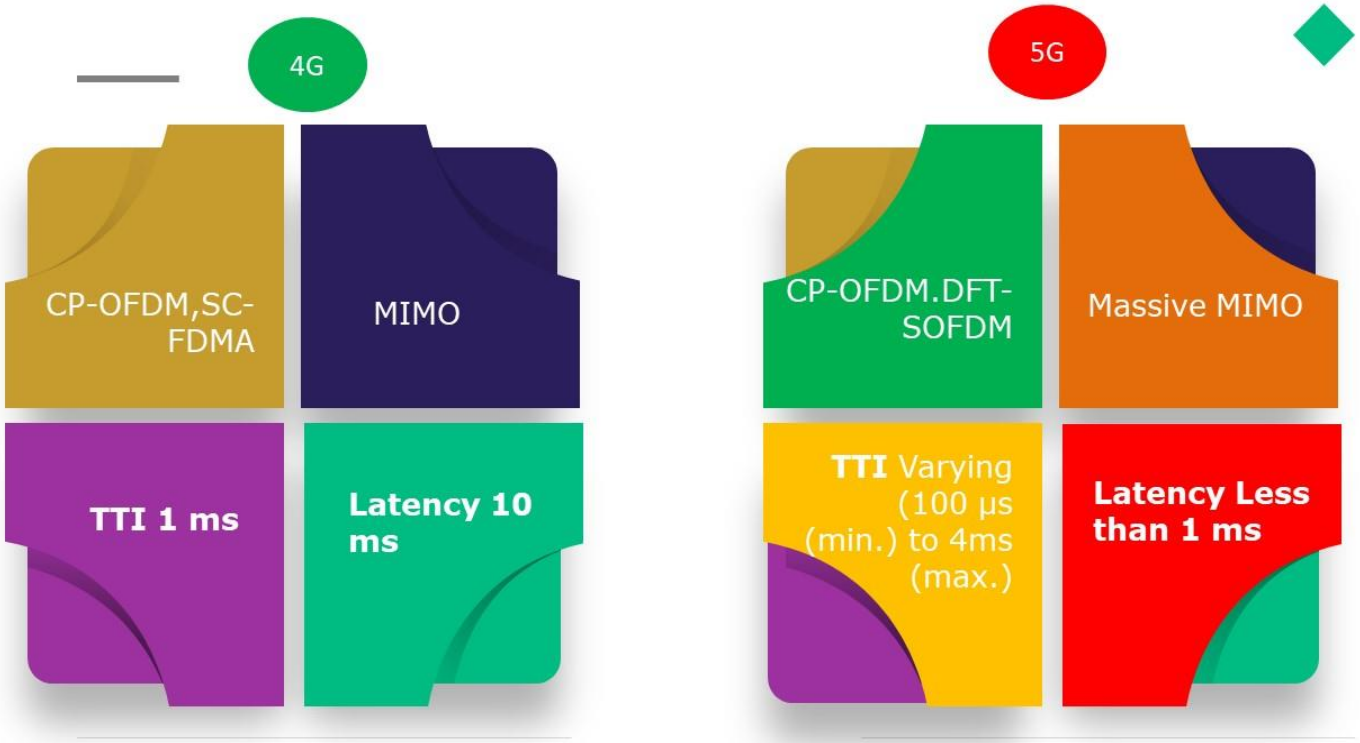
Ultra-reliable and Low Latency Communications (URLLC)

Latency < 1 ms

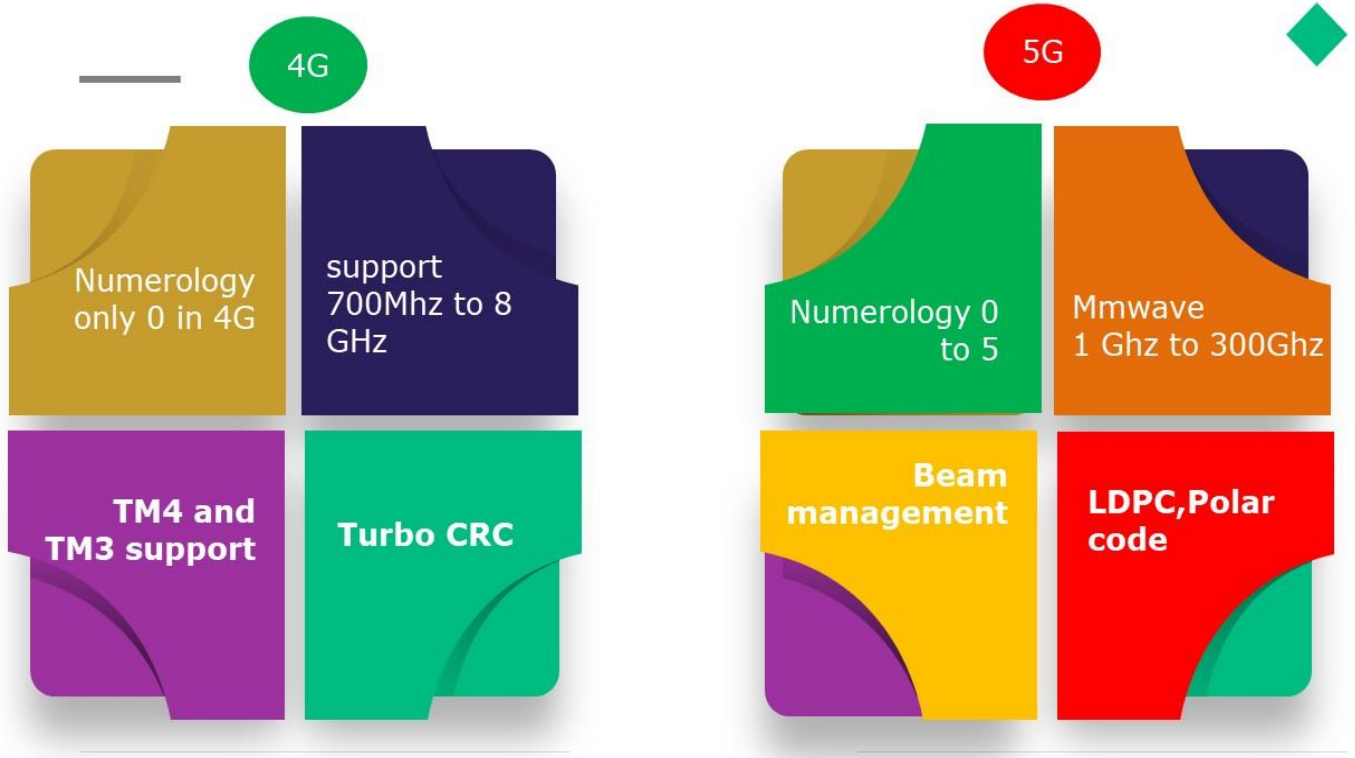
Comparing 4G and 5G



Comparison 4G LTE & 5G New Radio



Comparison 4G LTE & 5G New Radio



Comparison 4G LTE & 5G New Radio

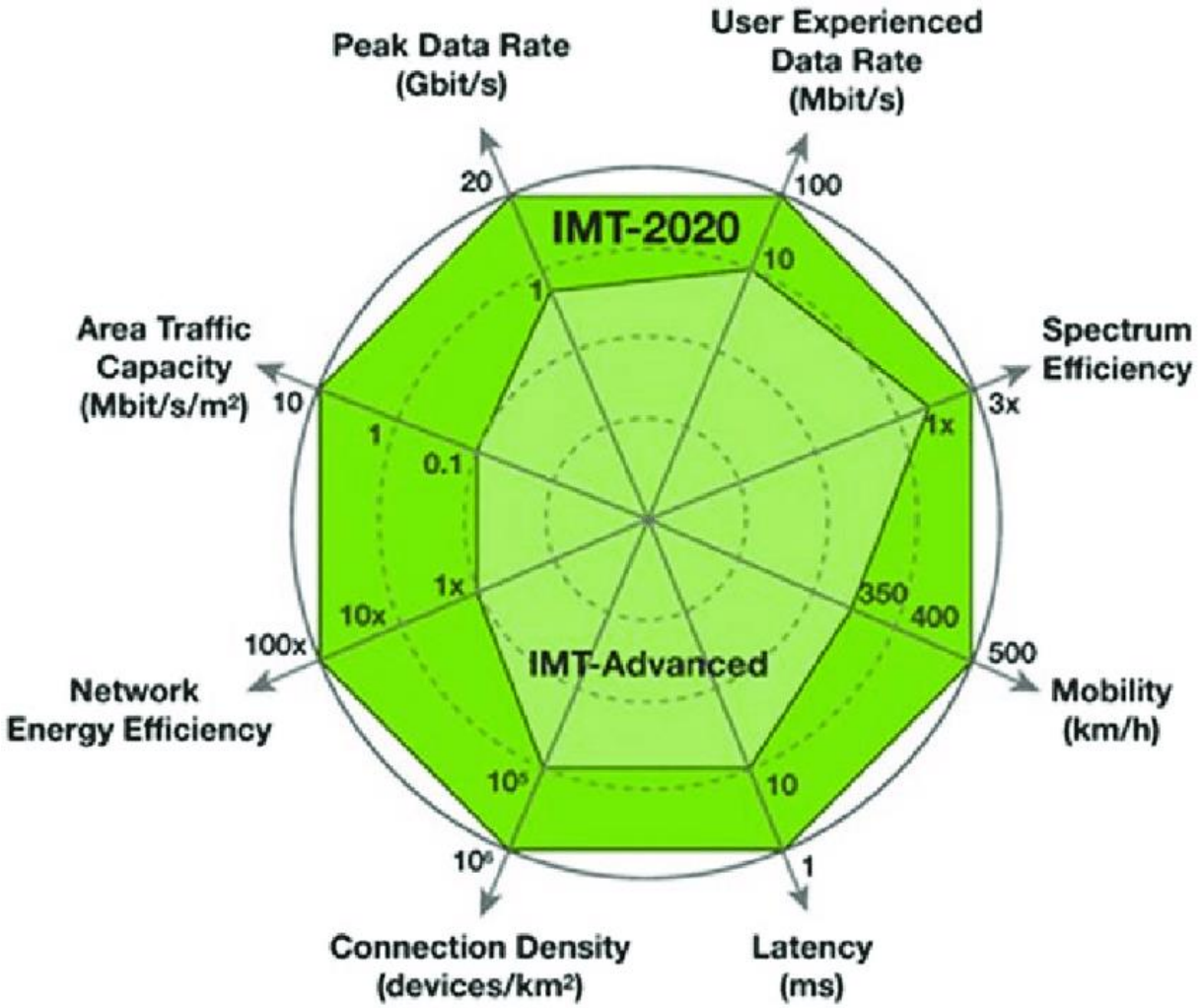


5G Targets and Standarization



ITU- R 5G Standard

The International Telecommunication Union Radio Communications Standardization Sector (ITU-R) has announced multi-fold design goals of 5G mobile networks known as International Mobile Telecommunications 2020 (IMT-2020), which include 20 Gb/s peak data rate, 100 Mb/s user experienced data rate, 10 Mb/s/m2 area traffic capacity, 10^6 devices/km2 connection density, 1 ms latency, mobility up to 500 km/h, backward compatibility to LTE/LTE-Advanced.



FLOATWAY SYSTEMS

3GPP RAN Requirements 38.913

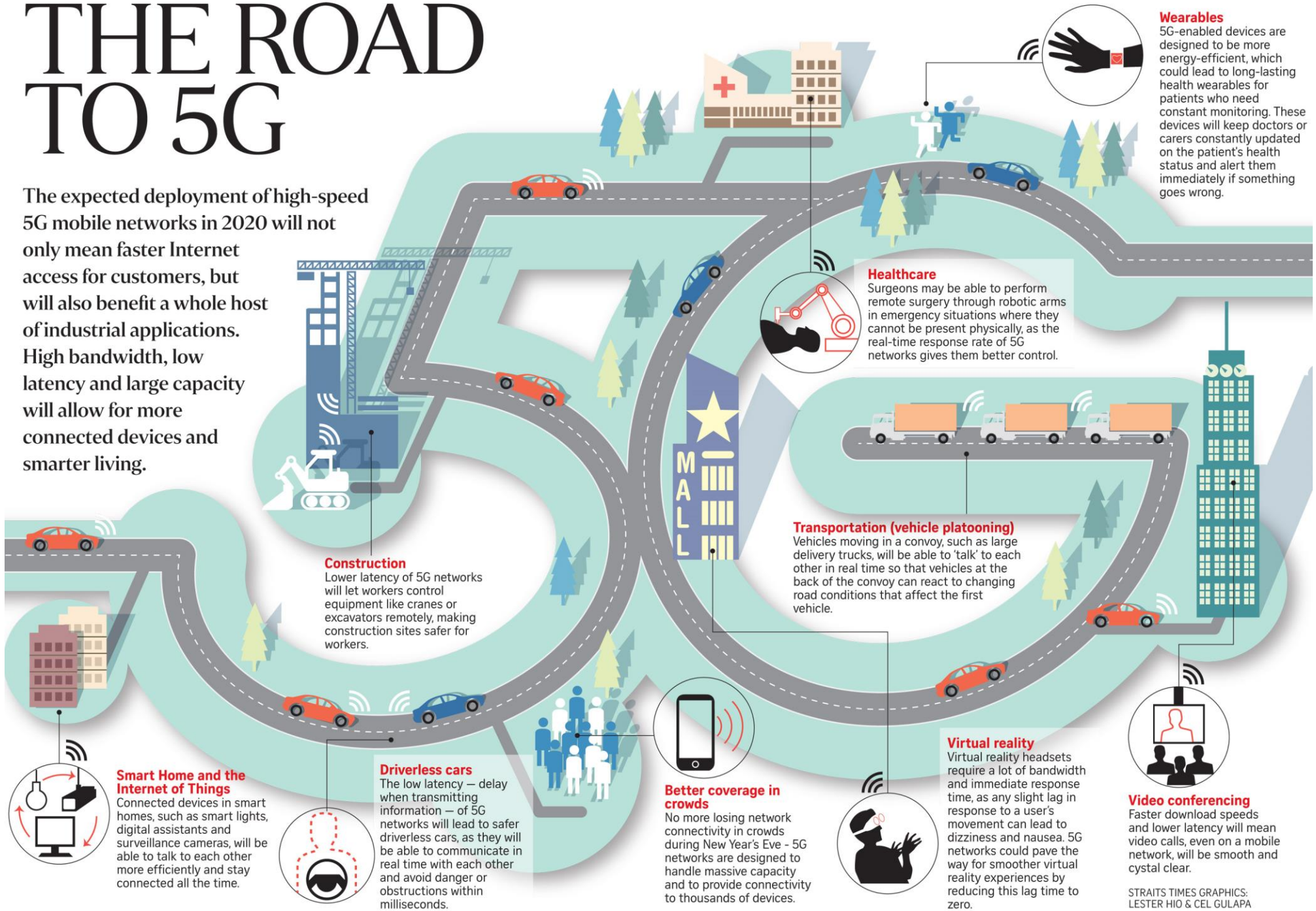
Performance Measure	Requirement
Peak data rate	DL: [20 Gbps] UL: [10 Gbps]
Peak spectral efficiency	DL: [30 bps/Hz] UL: [15 bps/Hz]
Spectrum Scalability	Yes
Bandwidth	Reference to IMT-2020
Bandwidth Scalability	Yes
Control plane latency	[10 ms]
UP latency URLLC, one-way	[0,5 ms]
UP latency eMBB, one way	[4ms]
Latency for infrequent small packets	10s / 20byte packet
Mobility interruption time (intra-syst.)	[0 ms]
Mobility	Up to 500 km/h
Inter-system mobility	Yes
Reliability	[1-10^{-5}] in [1ms]

Performance Measure	Requirement
UE Battery life	10-15 years
UE energy efficiency	Inspection (Qualitative)
Cell/Tx Point/TRP sp. Eff.	3xIMT-A requirement
Area traffic capacity	10Mbps/m^2 [ITU]
TRP spectral efficiency	[3x IMT-A requirement]
User experienced data rate	100/50 Mbps DL/UL [ITU]
User sp. eff. at 5% percentile	[3x cell edge IMT-A requirement]
Connection density	[1,000,000 devices/Km2]
NW energy efficiency	Qualitative & Quantitative KPI
eMBB Extreme coverage	140/143 dB loss MaxCL (2/1(DL))
IoT Coverage	MCL [164dB] for [160bps]
Support of wide range of services	Yes

5G Potential Use Case

THE ROAD TO 5G

The expected deployment of high-speed 5G mobile networks in 2020 will not only mean faster Internet access for customers, but will also benefit a whole host of industrial applications. High bandwidth, low latency and large capacity will allow for more connected devices and smarter living.



Wearables
5G-enabled devices are designed to be more energy-efficient, which could lead to long-lasting health wearables for patients who need constant monitoring. These devices will keep doctors or carers constantly updated on the patient's health status and alert them immediately if something goes wrong.

Healthcare
Surgeons may be able to perform remote surgery through robotic arms in emergency situations where they cannot be present physically, as the real-time response rate of 5G networks gives them better control.

Transportation (vehicle platooning)
Vehicles moving in a convoy, such as large delivery trucks, will be able to 'talk' to each other in real time so that vehicles at the back of the convoy can react to changing road conditions that affect the first vehicle.

Construction
Lower latency of 5G networks will let workers control equipment like cranes or excavators remotely, making construction sites safer for workers.

Smart Home and the Internet of Things
Connected devices in smart homes, such as smart lights, digital assistants and surveillance cameras, will be able to talk to each other more efficiently and stay connected all the time.

Driverless cars
The low latency — delay when transmitting information — of 5G networks will lead to safer driverless cars, as they will be able to communicate in real time with each other and avoid danger or obstructions within milliseconds.

Better coverage in crowds
No more losing network connectivity in crowds during New Year's Eve - 5G networks are designed to handle massive capacity and to provide connectivity to thousands of devices.

Virtual reality
Virtual reality headsets require a lot of bandwidth and immediate response time, as any slight lag in response to a user's movement can lead to dizziness and nausea. 5G networks could pave the way for smoother virtual reality experiences by reducing this lag time to zero.

Video conferencing
Faster download speeds and lower latency will mean video calls, even on a mobile network, will be smooth and cystal clear.

STRAITS TIMES GRAPHICS:
LESTER HIO & CEL GULAPA

5G Applications

1. **Autonomous Vehicle: The automatic controlled driving car and vehicles are key enablers of Vehicle-to-Vehicle (V2V), Vehicle-to-Infrastructure (V2I), and other Intelligent Transport Systems (ITS)**. The 5G network supports large bandwidth and low latency for these applications with high connection reliability. This network supports collision avoidance and intelligent navigation for the reliable transportation systems. uRLLC will support this feature in 5G.

2. **Public Safety Communications (PSCs): The PSCs facilitated various communication services in case of emergencies when the primary communication infrastructure is not available.** The PSCs incorporated rapid deployment and accessibility of communication setup. Therefore, 5G-enabled communications support a wide range of applications and long-term connectivity for the services in case of emergencies

3. **Military Applications: The military involved mission-critical control application which requires high data rate and long-term connectivity including the security parameters.** The real-time surveillance and monitoring of suspected areas require a network with the large bandwidth and low latency. The 5G-enabled network application is the best fit for such mission-oriented applications.

5G Applications

4. **Industrial Applications: Industrial automation composes massive IoT networks which require high connection density and low-power consumption that can be ensured through 5G setups.**

5. **e-Health Applications: The e-Health application requires remote diagnosis and long-term monitoring.** The e-Health setup involves video streaming embedded devices and advanced robotics which operate over the network that has low-power, low-latency, and high-throughput requirements. The 5G facilitate these requirements and serve as the best solution for these applications.

6. **Smart City Applications: Smart city adopts IoT devices, connected utilities, transportation, health care, education, smart grid, etc.** These applications scenario requires automation, cloud infrastructure, and artificial intelligence which operate over a network composing large bandwidth, high throughput, high connection density, and low latency. The 5G networks can also be adopted by new technologies and industry enablers like robotics and drones, etc.

Ultra-Reliable Low Latency Communication (URLLC) Applications

Scenario	End-to-end latency	Communication service availability	Reliability	User experienced data rate	Payload size	Traffic density	Connection density	Service area dimension
Discrete automation – motion control	1 ms	99,9999%	99,9999%	1 Mbps up to 10 Mbps	Small	1 Tbps/km²	100 000/km²	100 x 100 x 30 m
Discrete automation	10 ms	99,99%	99,99%	10 Mbps	Small to big	1 Tbps/km²	100 000/km²	1000 x 1000 x 30 m
Process automation – remote control	50 ms	99,9999%	99,9999%	1 Mbps up to 100 Mbps	Small to big	100 Gbps/km²	1 000/km²	300 x 300 x 50 m
Process automation – monitoring	50 ms	99,9%	99,9%	1 Mbps	Small	10 Gbps/km²	10 000/km²	300 x 300 x 50
Electricity distribution – medium voltage	25 ms	99,9%	99,9%	10 Mbps	Small to big	10 Gbps/km²	1 000/km²	100 km along power line
Electricity distribution – high voltage	5 ms	99,9999%	99,9999%	10 Mbps	Small	100 Gbps/km²	1 000/km²	200 km along power line
Intelligent transport systems – infrastructure backhaul	10 ms	99,9999%	99,9999%	10 Mbps	Small to big	10 Gbps/km²	1 000/km²	2 km along a road
Tactile interaction	0,5 ms	[99,999%]	[99,999%]	[Low]	[Small]	[Low]	[Low]	TBC
Remote control	[5 ms]	[99,999%]	[99,999%]	[From low to 10 Mbps]	[Small to big]	[Low]	[Low]	TBC

Discrete automation – Discrete automation is characterized by high requirements on the communications system regarding reliability and availability. Systems supporting discrete automation are usually deployed in geographically limited areas, access to them may be limited to authorized users, and they may be isolated from networks or network resources used by other cellular customers.

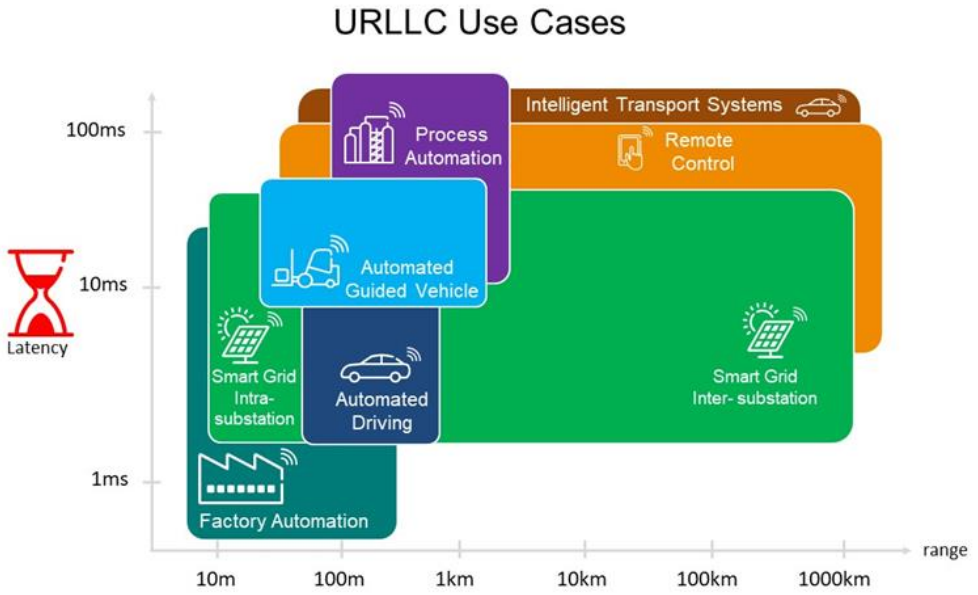
Process automation – Automation for (reactive) flows, e.g., refineries and water distribution networks. Process automation is characterized by high requirements on the communications system regarding communication service availability. Systems supporting process automation are usually deployed in geographically limited areas, access to them is usually limited to authorized users, and it will usually be served by private networks.

Automation for electricity distribution (mainly medium and high voltage). Electricity distribution is characterized by high requirements on the communications service availability. In contrast to the above use cases, electricity distribution is deeply immersed into the public space. Since electricity distribution is an essential infrastructure, it will, as a rule, be served by private networks.

Intelligent transport systems – Automation solutions for the infrastructure supporting street-based traffic. This use case addresses the connection of the road-side infrastructure, e.g., road side units, with other infrastructure, e.g., a traffic guidance system. As is the case for automation electricity, the nodes are deeply immersed into the public space.

Ultra-Reliable Low Latency Communication (URLLC) Applications

Industry Vertical	Application
Smart Factory/Industrial Automation	Industrial Control Robot Control Machine to Machine Process Control
Healthcare Industry	Remote Diagnosis Emergency Response Remote Surgery
Entertainment Industry	Immersive Entertainment Online Gaming
Transport Industry	Driver Assistance Applications Enhanced Safety Autonomous Driving Traffic Management
Manufacturing Industry	Motion Control Remote Control AR and VR Applications
Energy Sector	Smart Energy Smart Grid



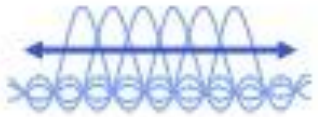
5G New Radio & Features



3GPP Rel-15 establishes a solid foundation for 5G NR

For enhanced mobile broadband and beyond

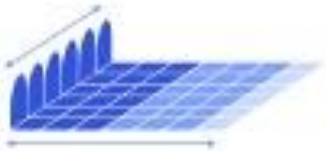
Scalable OFDM-based air interface



Scalable OFDM numerology

Efficiently address diverse spectrum, deployments/services

Flexible slot-based framework



Self-contained slot structure

Key enabler to low latency, URLLC and forward compatibility

Advanced channel coding



Multi-Edge LDPC and CRC-Aided Polar

Efficiently support large data blocks and a reliable control channel

Massive MIMO



Reciprocity-based MU-MIMO

Efficiently utilize a large number of antennas to increase coverage/capacity

Mobile mmWave



Beamforming and beam-tracking

Enables wide mmWave bandwidths for extreme capacity and throughput

Qualcomm

Our technology inventions are driving Rel-15 specifications

Early R&D investments | Best-in-class prototypes | Fundamental contributions to 3GPP

1. Scalable OFDM-based Air Interface

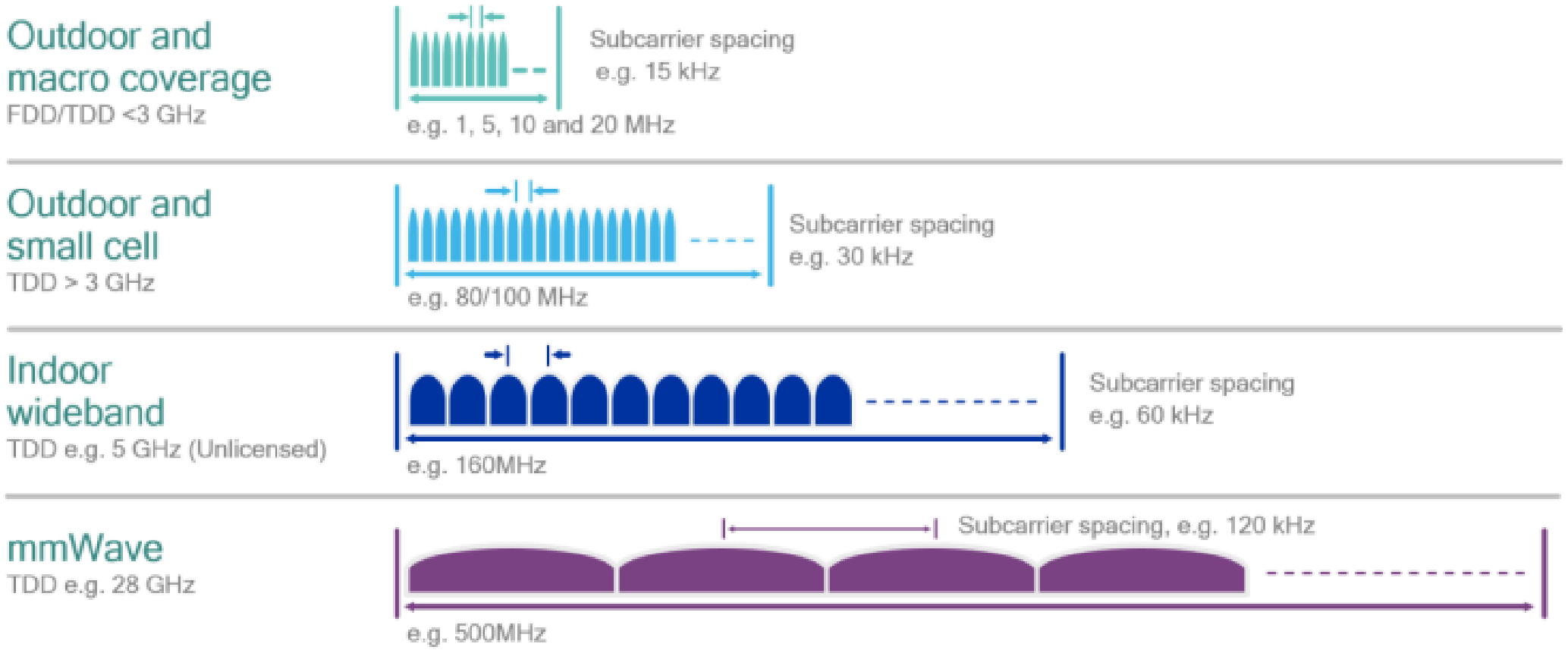
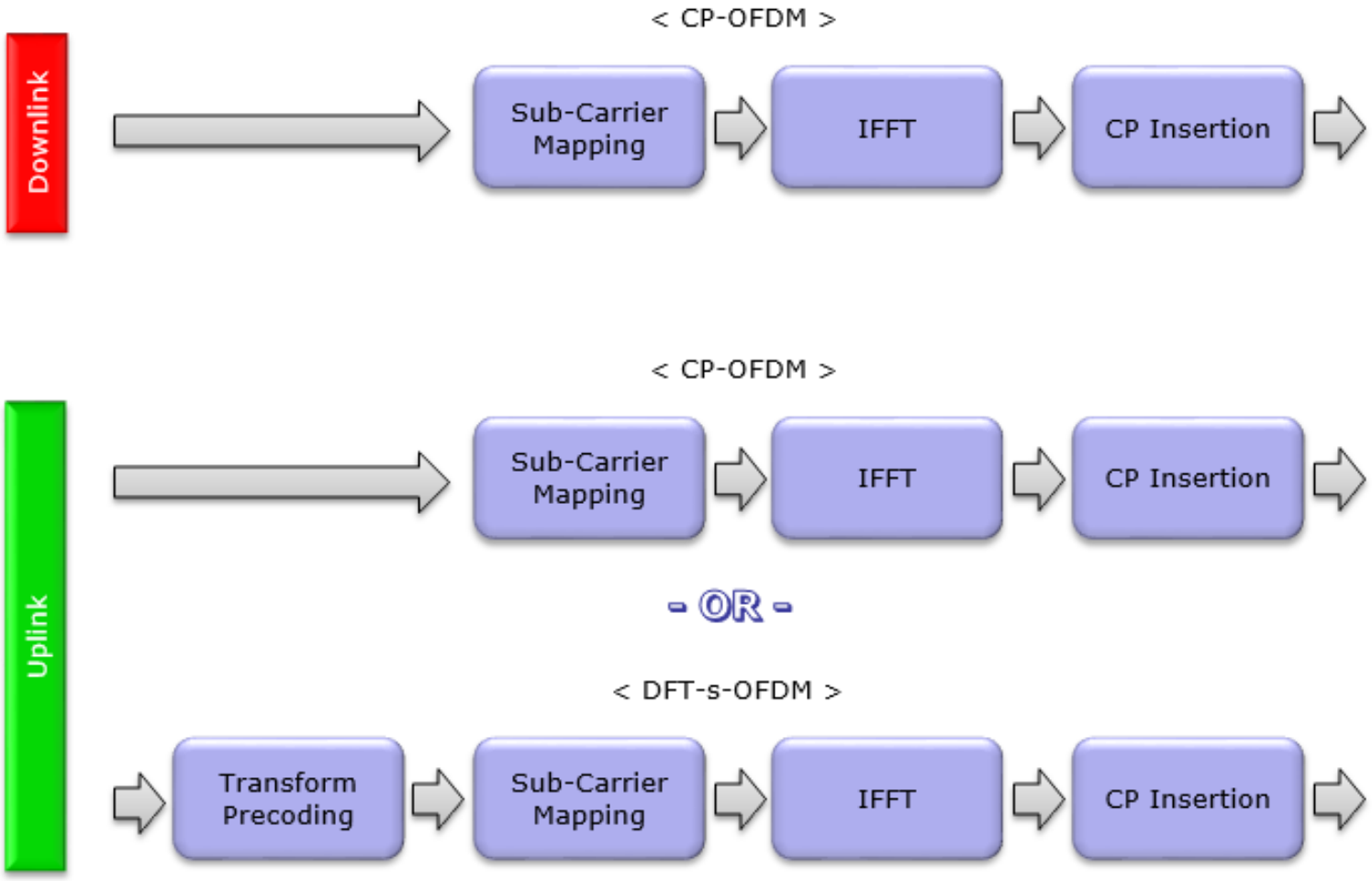


Figure 10: Example usage models, channel bandwidths, and subcarrier spacing

1.1 OFDM Waveform

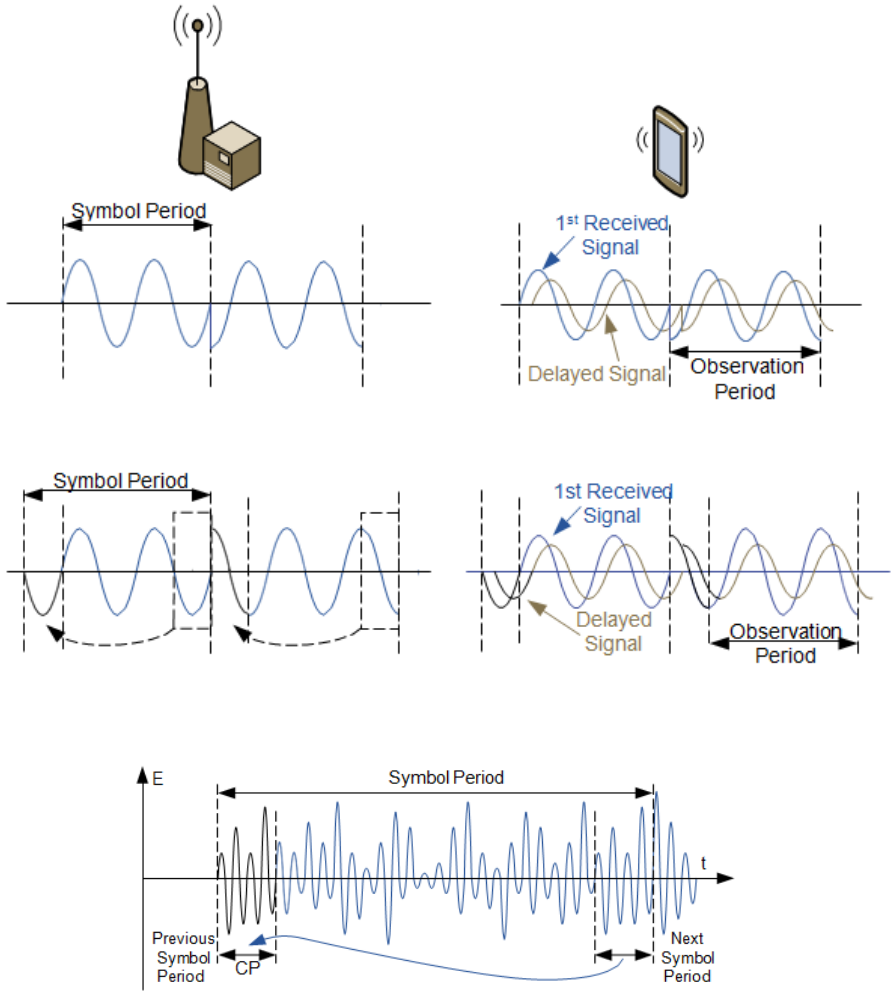


Unlike LTE, NR has two options for UL Waveform. One is CP-OFDM (same as DL Waveform) and the other one is DFT-s-OFDM which is same as LTE UL waveform.

1.2 Cyclic Prefix

In most OFDM systems, a CP (Cyclic Prefix) can be used to counter the effects of multipath propagation. This can be represented as a “guard period” which is located just in front of the user data portion and as such, is able to mitigate Delay Spread. The CP is in fact the final part of the OFDM Symbol copied and placed in front of the same OFDM symbol.

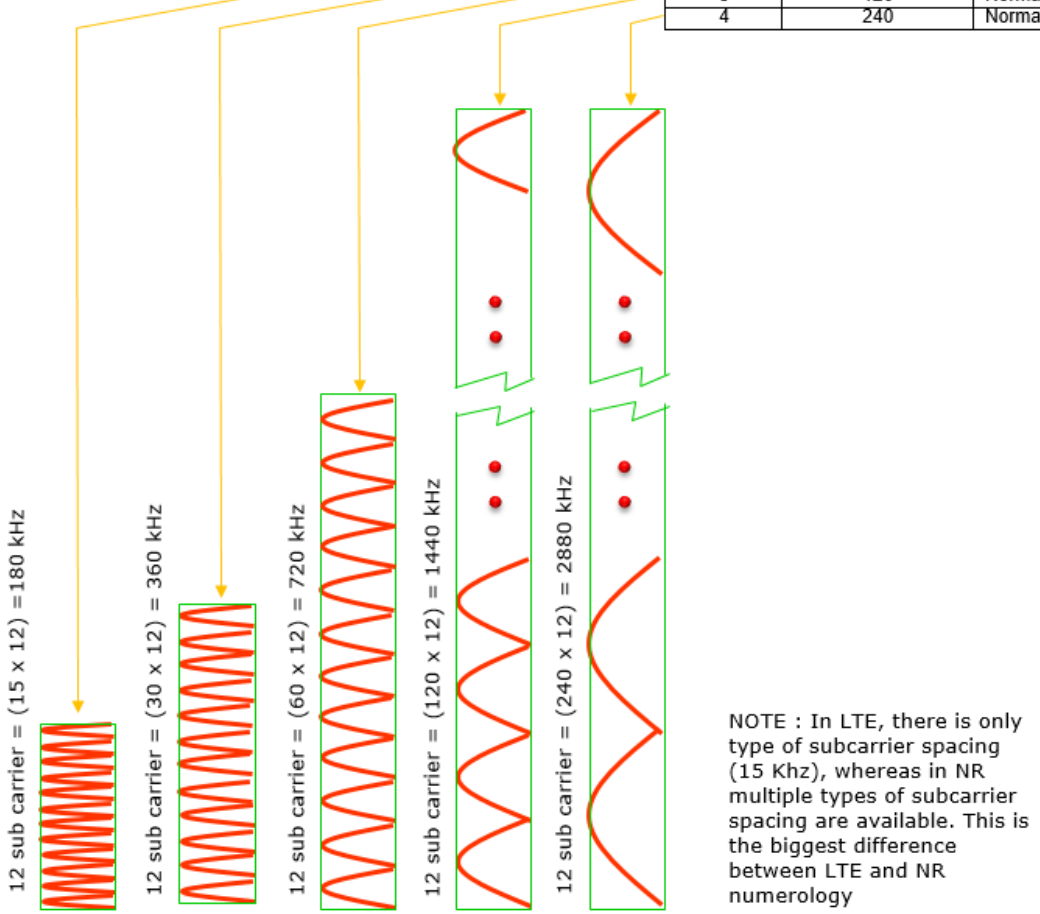
In this example the CP represents 20% of the symbol and since the delay spread in this case was only about 10% the symbol can be received. For 5G NR there is a normal CP and an extended CP. The extended CP provides a greater period for multipath protection and hence is linked to large cell sizes. Unfortunately, extended CP also reduces the number of symbols per slot/subframe and as such the data rate is reduced.



1.3 Numerology

38.211 Table 4.2-1

μ	$\Delta f = 2^{\mu} \cdot 15$ [kHz]	Cyclic prefix
0	15	Normal
1	30	Normal
2	60	Normal, Extended
3	120	Normal
4	240	Normal

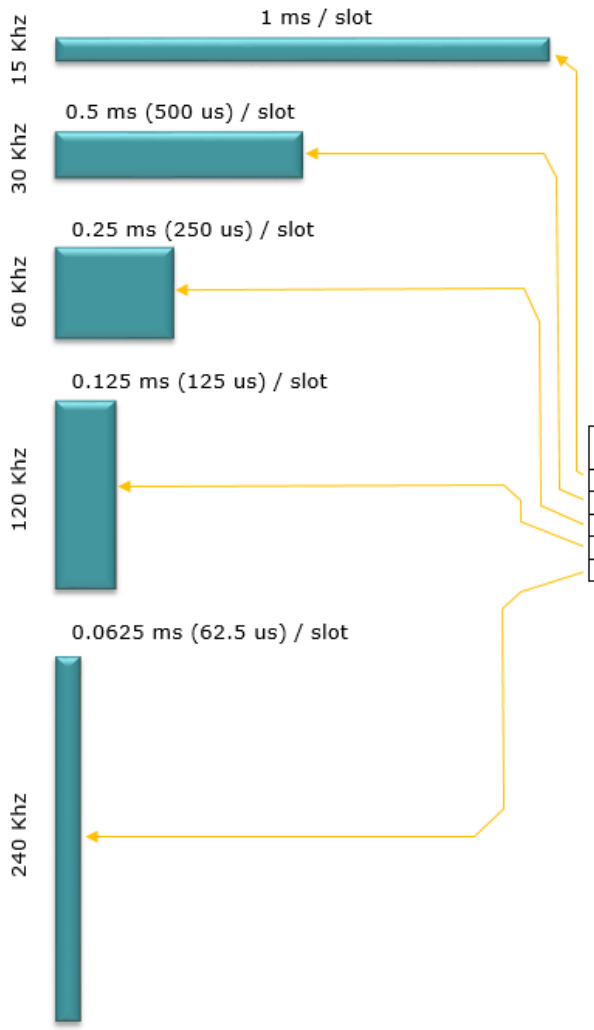


NOTE : In LTE, there is only type of subcarrier spacing (15 Khz), whereas in NR multiple types of subcarrier spacing are available. This is the biggest difference between LTE and NR numerology

In NR, transmitters and receivers may enjoy a wider bandwidth at high-frequency bands. In this case, the subcarrier spacing can be extended (larger than 15 kHz as adopted by LTE/LTE-A, and potentially up to 480 kHz).

In addition, high carrier frequencies are also vulnerable to the Doppler Effect, and a large subcarrier spacing may facilitate inter-carrier interference (ICI) mitigation.

1.4 Numerology & Slot Lenght



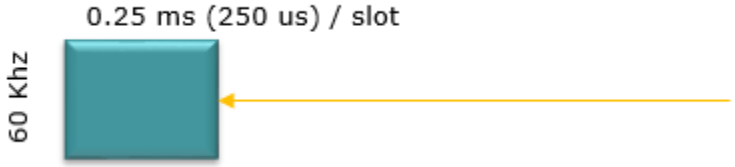
Normal Cyclic Prefix

< 38.211 – Table 4.3.2-1 >

μ	N^{slot}_{symb}	$N^{frame,\mu}_{slot}$	$N^{subframe,\mu}_{slot}$
0	14	10	1
1	14	20	2
2	14	40	4
3	14	80	8
4	14	160	16

The general tendency is that slot length gets shorter as subcarrier spacing gets wider. Actually this tendency comes from the nature of OFDM.

Extended Cyclic Prefix

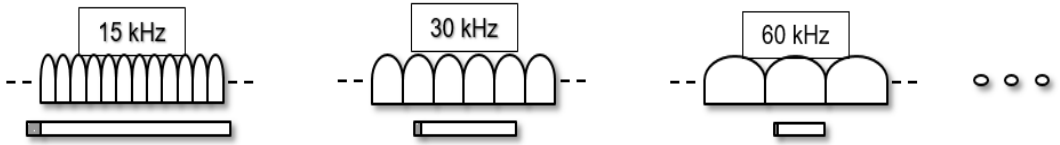


< 38.211 – v2.0.0 Table 4.3.2-w >

μ	N^{slot}_{symb}	$N^{frame,\mu}_{slot}$	$N^{subframe,\mu}_{slot}$
2	12	40	4

1.4 Numerology & Slot Lenght

SCS KHZ	Slot configuration		
	Number of Symbols/Slot	Number of Slots/Subframe	Number of Slots /Frame
15	14	1	10
30	14	2	20
60	14	4	40
120	14	8	80
240	14	16	160

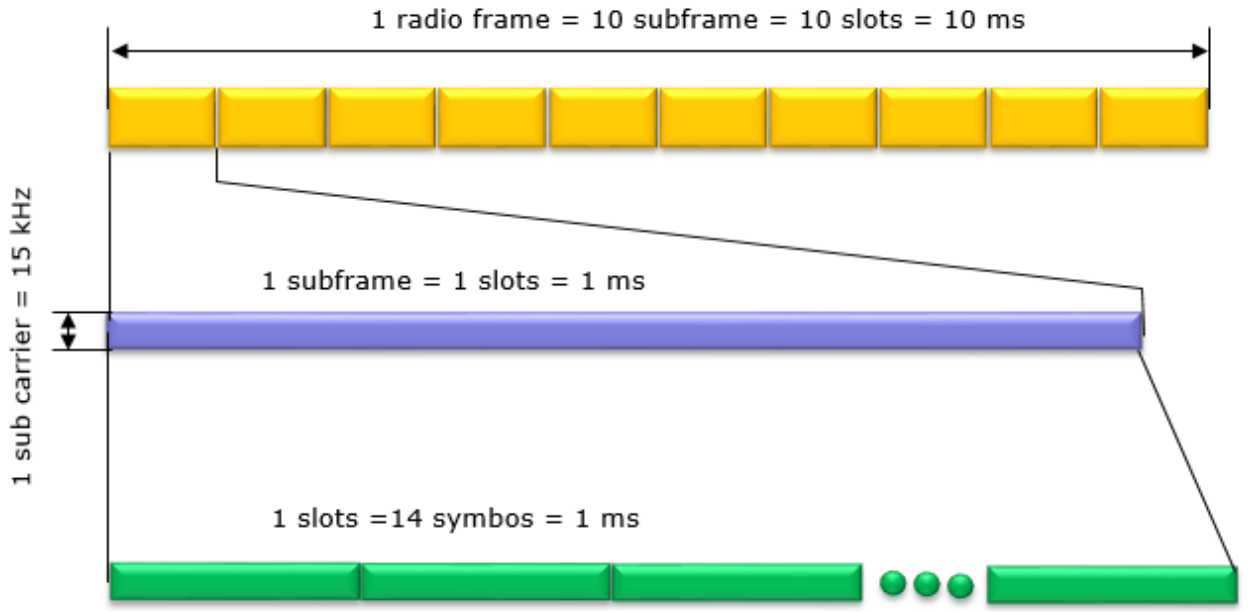


	Normal CP / L freq.	Opt. CP / L freq. / L latency	Small CP / H freq. / L latency	Extended CP
Subcarrier BW	15 kHz	30 kHz (2x15 kHz)	60 kHz (4x15 kHz)	$(n \times 15\ \text{kHz}, n = 1,2,4)$
SF duration	500 µs	250 µs	125 µs	500/n µs
OFDM symbol, duration	66.67 µs	33.33 µs	16.67 µs	66.67/n µs
CP, duration	4.76 µs	2.38 µs	1.19 µs	16.67/n µs
OFDM symbol incl. CP	71.43 µs	35.71 µs	17.86 µs	83.33/n µs

1.5 Normal Cyclic Prefix – Numerology 0 (15 kHz)

< 38.211 – Table 4.3.2-1 >

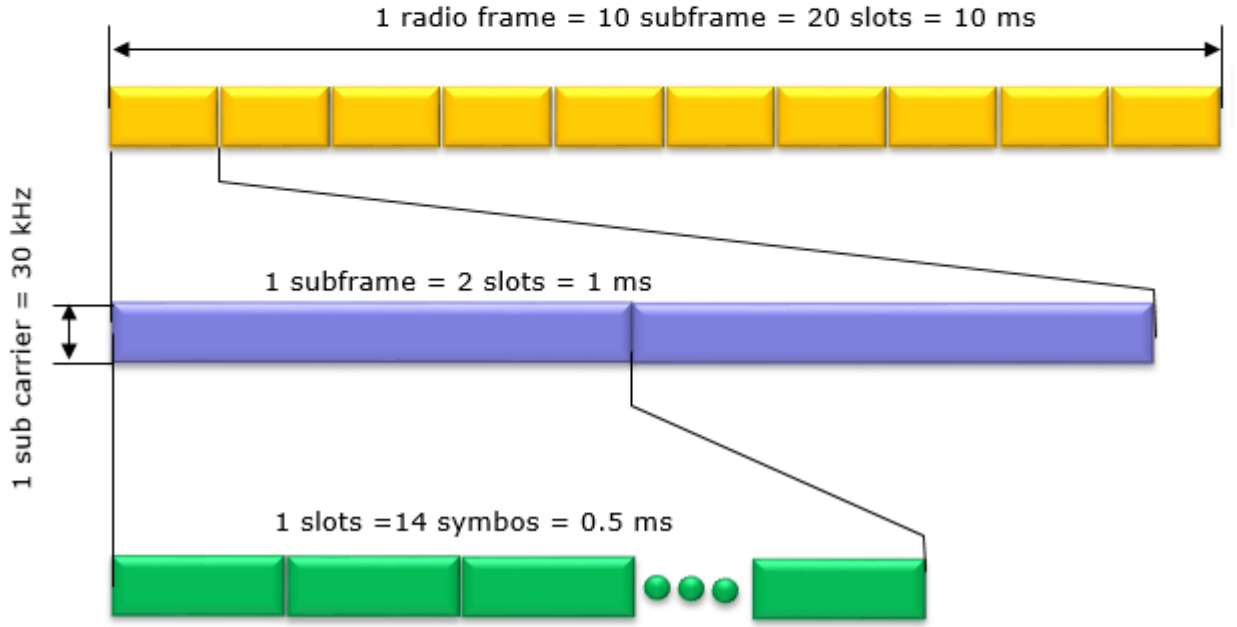
μ	N^{slot}_{symb}	$N^{frame,\mu}_{slot}$	$N^{subframe,\mu}_{slot}$
0	14	10	1
1	14	20	2
2	14	40	4
3	14	80	8
4	14	160	16



1.6 Normal Cyclic Prefix – Numerology 1 (30 kHz)

< 38.211 – Table 4.3.2-1 >

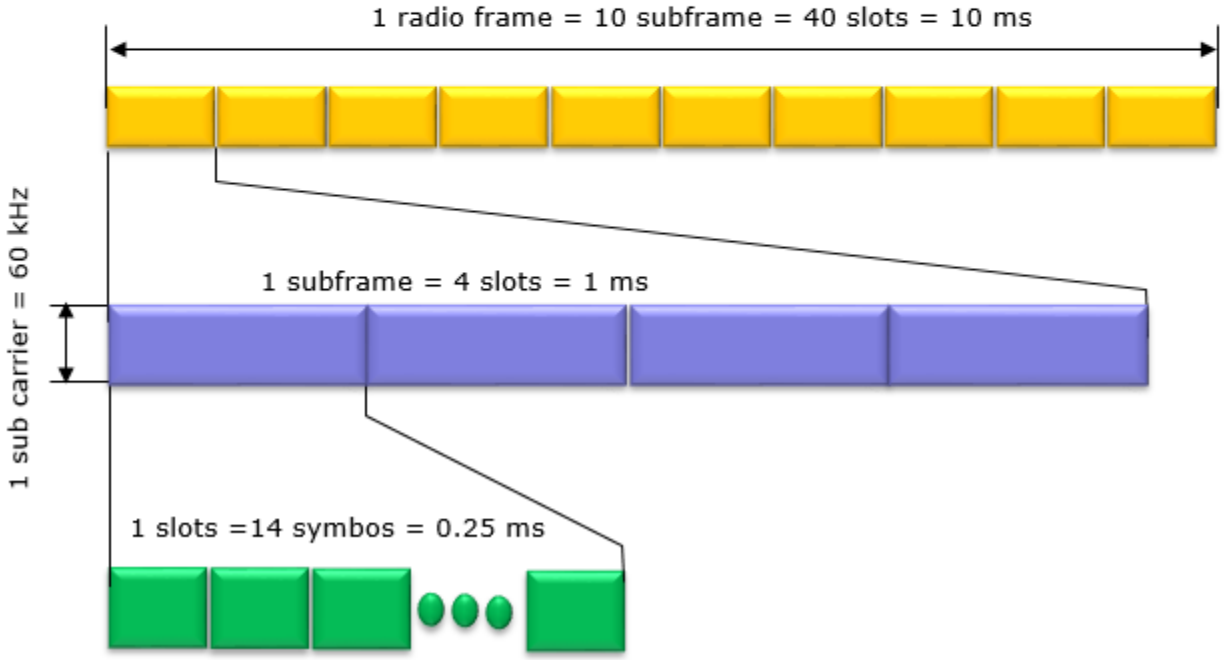
μ	N^{slot}_{symb}	$N^{frame,\mu}_{slot}$	$N^{subframe,\mu}_{slot}$
0	14	10	1
1	14	20	2
2	14	40	4
3	14	80	8
4	14	160	16



1.7 Normal Cyclic Prefix – Numerology 2 (60 kHz)

< 38.211 – Table 4.3.2-1 >

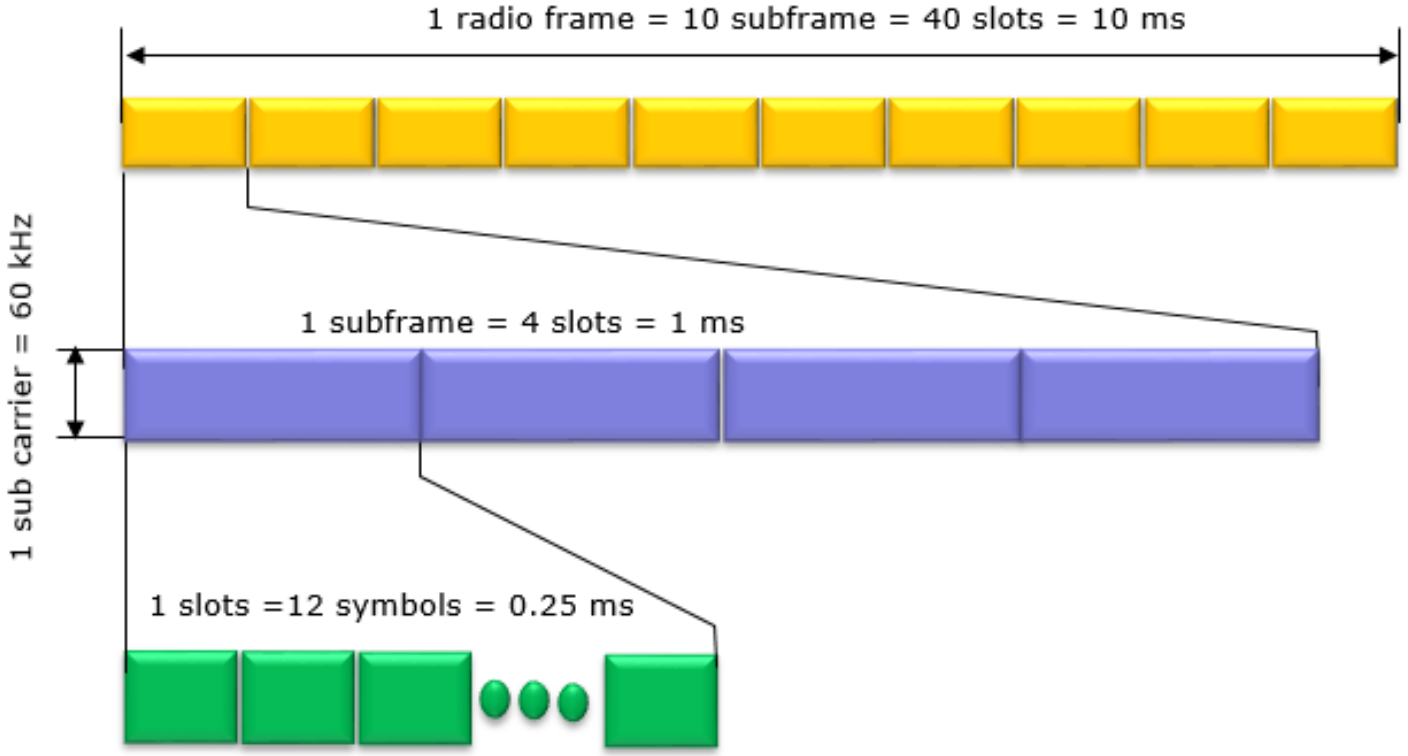
μ	N^{slot}_{symb}	$N^{frame,\mu}_{slot}$	$N^{subframe,\mu}_{slot}$
0	14	10	1
1	14	20	2
2	14	40	4
3	14	80	8
4	14	160	16



1.8 Extended Cyclic Prefix – Numerology 2 (60 kHz)

< 38.211 – v2.0.0 Table 4.3.2-w >

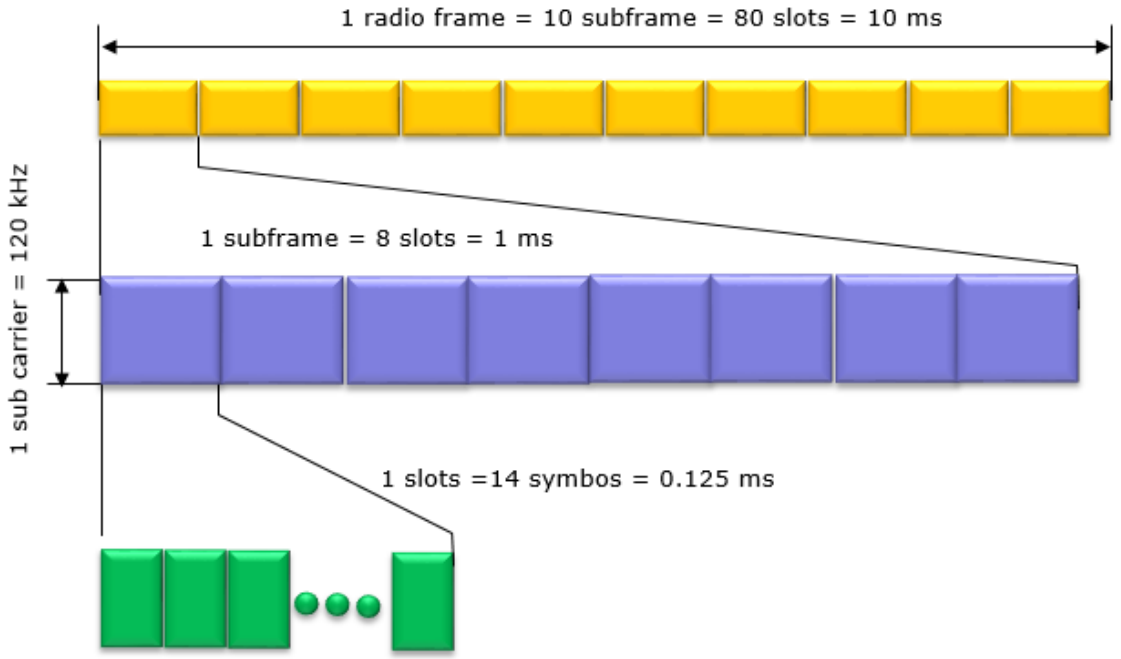
μ	N^{slot}_{symb}	$N^{frame,\mu}_{slot}$	$N^{subframe,\mu}_{slot}$
2	12	40	4



1.9 Normal Cyclic Prefix – Numerology 3 (120 kHz)

< 38.211 – Table 4.3.2-1 >

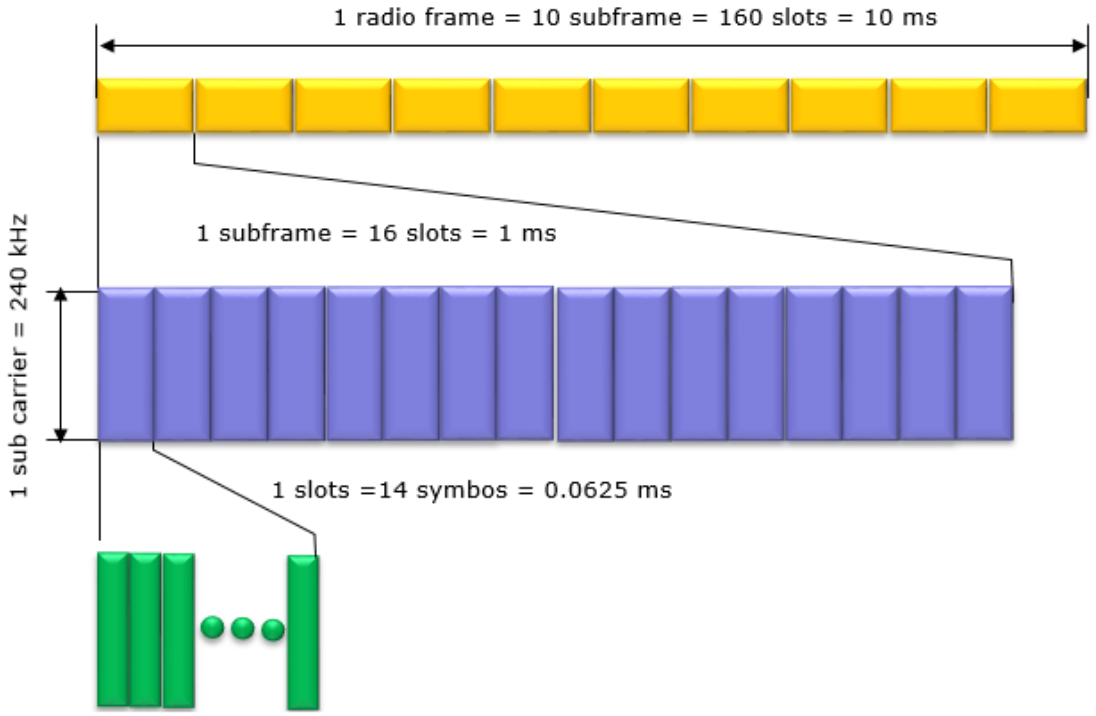
μ	N^{slot}_{symb}	$N^{frame,\mu}_{slot}$	$N^{subframe,\mu}_{slot}$
0	14	10	1
1	14	20	2
2	14	40	4
3	14	80	8
4	14	160	16



1.10 Normal Cyclic Prefix – Numerology 4 (240 kHz)

< 38.211 – Table 4.3.2-1 >

μ	N^{slot}_{symb}	$N^{frame,\mu}_{slot}$	$N^{subframe,\mu}_{slot}$
0	14	10	1
1	14	20	2
2	14	40	4
3	14	80	8
4	14	160	16



FLOATWAY SYSTEMS

1.11 Slot Format

	Symbol Number in a slot													
Format	0	1	2	3	4	5	6	7	8	9	10	11	12	13
0	D	D	D	D	D	D	D	D	D	D	D	D	D	D
1	U	U	U	U	U	U	U	U	U	U	U	U	U	U
2	F	F	F	F	F	F	F	F	F	F	F	F	F	F
3	D	D	D	D	D	D	D	D	D	D	D	D	D	F
4	D	D	D	D	D	D	D	D	D	D	D	D	F	F
5	D	D	D	D	D	D	D	D	D	D	D	F	F	F
6	D	D	D	D	D	D	D	D	D	D	F	F	F	F
7	D	D	D	D	D	D	D	D	D	F	F	F	F	F
8	F	F	F	F	F	F	F	F	F	F	F	F	F	U
9	F	F	F	F	F	F	F	F	F	F	F	F	U	U
10	F	U	U	U	U	U	U	U	U	U	U	U	U	U
11	F	F	U	U	U	U	U	U	U	U	U	U	U	U
12	F	F	F	U	U	U	U	U	U	U	U	U	U	U
13	F	F	F	F	U	U	U	U	U	U	U	U	U	U
14	F	F	F	F	F	U	U	U	U	U	U	U	U	U
15	F	F	F	F	F	F	U	U	U	U	U	U	U	U
16	D	F	F	F	F	F	F	F	F	F	F	F	F	F
17	D	D	F	F	F	F	F	F	F	F	F	F	F	F
18	D	D	D	F	F	F	F	F	F	F	F	F	F	F
19	D	F	F	F	F	F	F	F	F	F	F	F	F	U
20	D	D	F	F	F	F	F	F	F	F	F	F	F	U
21	D	D	D	F	F	F	F	F	F	F	F	F	F	U
22	D	F	F	F	F	F	F	F	F	F	F	F	U	U
23	D	D	F	F	F	F	F	F	F	F	F	F	U	U
24	D	D	D	F	F	F	F	F	F	F	F	F	U	U
25	D	F	F	F	F	F	F	F	F	F	F	U	U	U
26	D	D	F	F	F	F	F	F	F	F	F	U	U	U
27	D	D	D	F	F	F	F	F	F	F	F	U	U	U
28	D	D	D	D	D	D	D	D	D	D	D	D	F	U
29	D	D	D	D	D	D	D	D	D	D	D	F	F	U
30	D	D	D	D	D	D	D	D	D	D	F	F	F	U

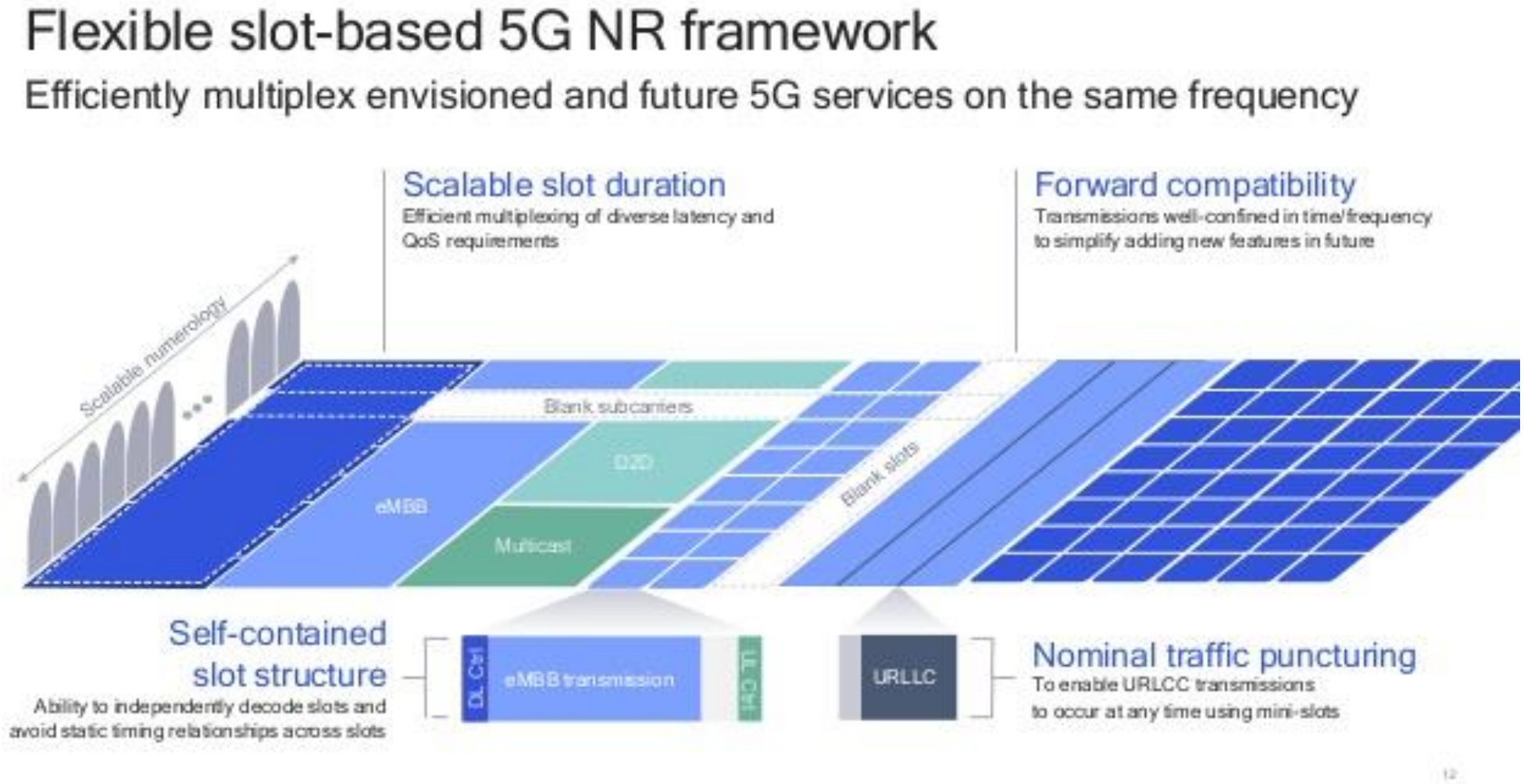
	Symbol Number in a slot													
Format	0	1	2	3	4	5	6	7	8	9	10	11	12	13
31	D	D	D	D	D	D	D	D	D	D	D	F	U	U
32	D	D	D	D	D	D	D	D	D	D	F	F	U	U
33	D	D	D	D	D	D	D	D	D	F	F	F	U	U
34	D	F	U	U	U	U	U	U	U	U	U	U	U	U
35	D	D	F	U	U	U	U	U	U	U	U	U	U	U
36	D	D	D	F	U	U	U	U	U	U	U	U	U	U
37	D	F	F	U	U	U	U	U	U	U	U	U	U	U
38	D	D	F	F	U	U	U	U	U	U	U	U	U	U
39	D	D	D	F	F	U	U	U	U	U	U	U	U	U
40	D	F	F	F	U	U	U	U	U	U	U	U	U	U
41	D	D	F	F	F	U	U	U	U	U	U	U	U	U
42	D	D	D	F	F	F	U	U	U	U	U	U	U	U
43	D	D	D	D	D	D	D	D	D	F	F	F	F	U
44	D	D	D	D	D	D	F	F	F	F	F	F	U	U
45	D	D	D	D	D	D	F	F	U	U	U	U	U	U
46	D	D	D	D	D	D	F	D	D	D	D	D	D	F
47	D	D	D	D	D	F	F	D	D	D	D	D	F	F
48	D	D	F	F	F	F	F	D	D	F	F	F	F	F
49	D	F	F	F	F	F	F	D	F	F	F	F	F	F
50	F	U	U	U	U	U	U		U	U	U	U	U	U
51	F	F	U	U	U	U	U	F	F	U	U	U	U	U
52	F	F	F	U	U	U	U	F	F	F	U	U	U	U
53	F	F	F	F	U	U	U	F	F	F	F	U	U	U
54	D	D	D	D	D	F	U	D	D	D	D	D	F	U
55	D	D	F	U	U	U	U	U	U	FU	U	U	U	U
56	D	F	U	U	U	U	U	D	F	U	U	U	U	U
57	D	D	D	D	F	F	U	D	D	D	D	F	F	U
58	D	D	F	F	U	U	U	D	D	F	F	U	U	U
59	D	F	U	U	U	U	U	D	F	U	U	U	U	U
60	D	F	F	F	F	F	U	D	F	F	F	F	F	U
61	D	D	F	F	F	F	U	D	D	F	F	F	F	U
62-255														

In LTE TDD, if a subframe (equivalent to a Slot in NR) is configured for DL or UL, all of the symbols within the subframe should be used as DL or UL.

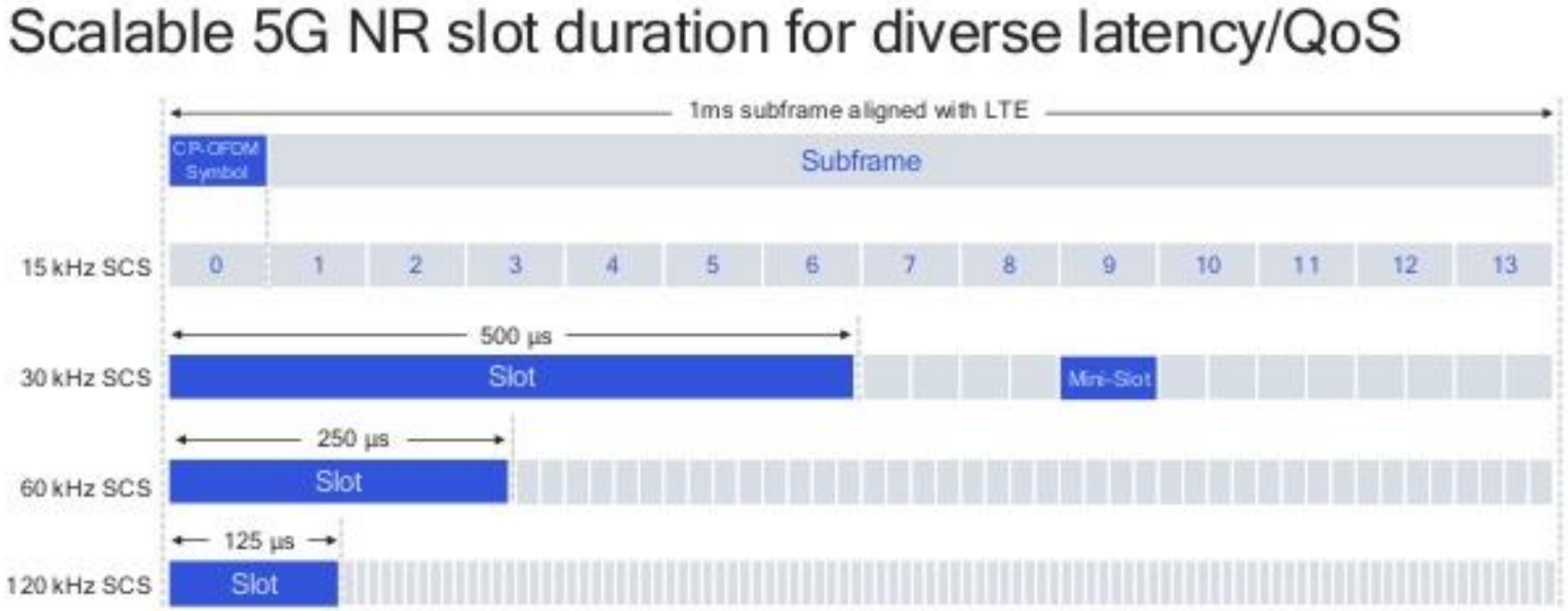
But in NR, the symbols within a slot can be configured in various ways as follows

D = Downlink
U = Uplink
F = Flexible

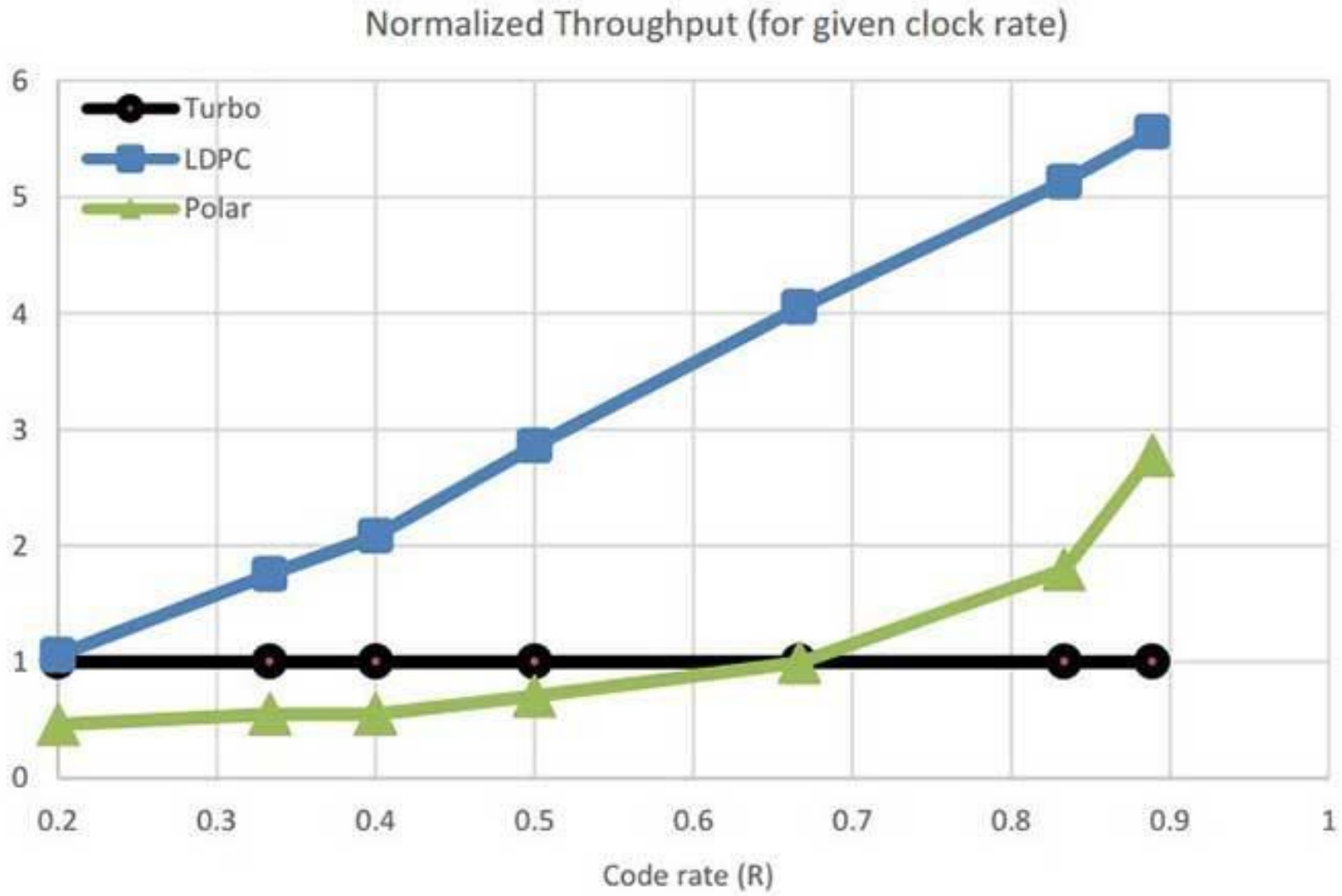
2. Flexible Slot-based Framework



2. Flexible Slot-based Framework



3. Advanced Channel Coding



4. Massive MIMO

5G NR Massive MIMO

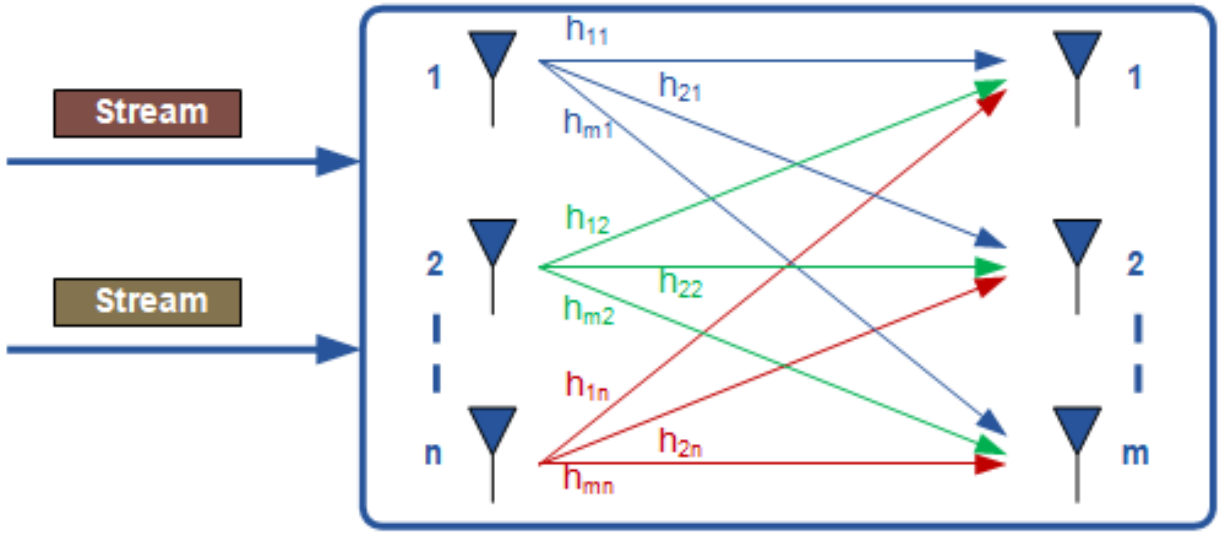
- From 2/4/8 to massive number of antennas, e.g., 256 or 1024
- mmWave antennas with smaller antenna size
- Main benefits
  - Capacity gains
  - Spectral efficiency
  - Energy efficiency
- Support up to 8 layers for SU-MIMO and up to 12 layers for MU-MIMO
- More accurate CSI feedback: type I and type II CSI
- UL supports both CB and NCB





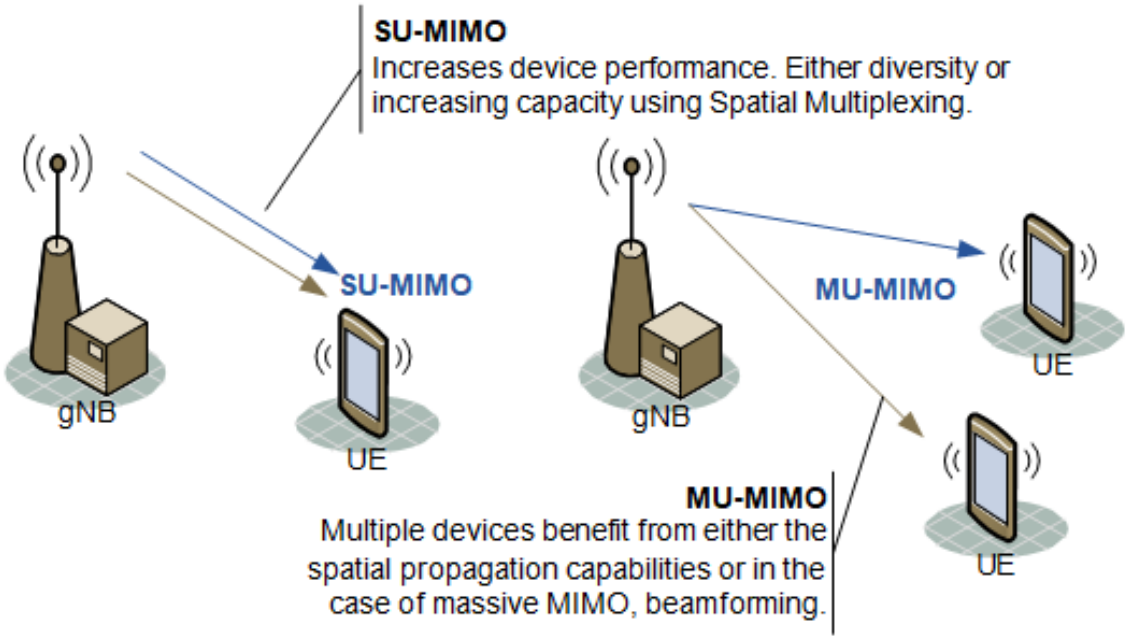
4.1 MIMO Spatial Multiplexing

MIMO systems can identify separate signals, i.e. layers/streams and these can be used to carry the same information, effectively a form of diversity. However, they can also be used to carry different information, which is termed SM (Spatial Multiplexing) and typically requires good radio conditions, i.e. a spatially diverse channel with good signal levels. Spatial Multiplexing implies “different” data on the streams.



4.2 SU-MIMO vs MU-MIMO

Like LTE, 5G NR supports both SU-MIMO (Single-User MIMO) and MU-MIMO (Multi-User MIMO). SU-MIMO, as it name suggests is designed to increase the performance for a single user. In contrast, MU-MIMO is designed to serve multiple users simultaneously on the same frequency resource, primarily relying on spatial separation or beam separation.



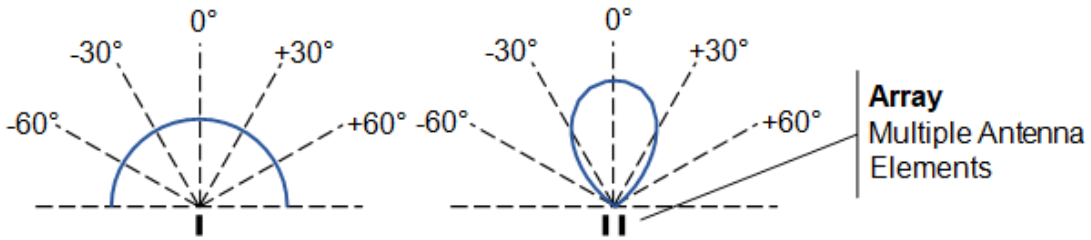
4.3 Spatial Multiplexing vs Beamforming

These are not mutually exclusive, however typically in the sub 6GHz band the channel characteristics are rich in multipaths and therefore ideally suited for spatial multiplexing. In contrast, at mmWave the system relies on beamforming (to get the power to the device) which has fewer multipaths.

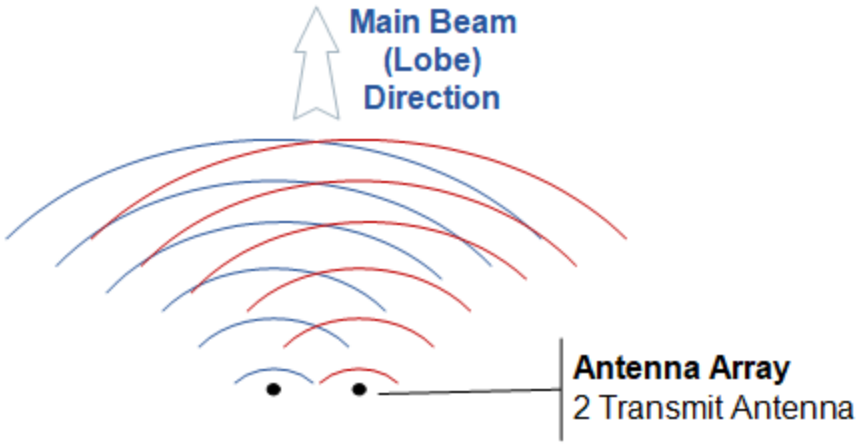
	Sub 6GHz	mmWave
MIMO Order	Up to 8x8	Less MIMO (usually 2x2)
Main Technique	Spatial Multiplexing	Beamforming to a Single Device
Channel Characteristics	Rich Multipaths ideal for Spatial Multiplexing	Fewer Multipaths due to beamforming

4.4 Creating a Beam

The easiest method to create a beam is to place multiple antennas in an “array”. Note that there many implementation options when it comes to aligning the antenna elements. The simplest is illustrated besides whereby the antennas are aligned in a line. Notice that as an additional antenna element is added, the radiation pattern is changed, becoming more directional.

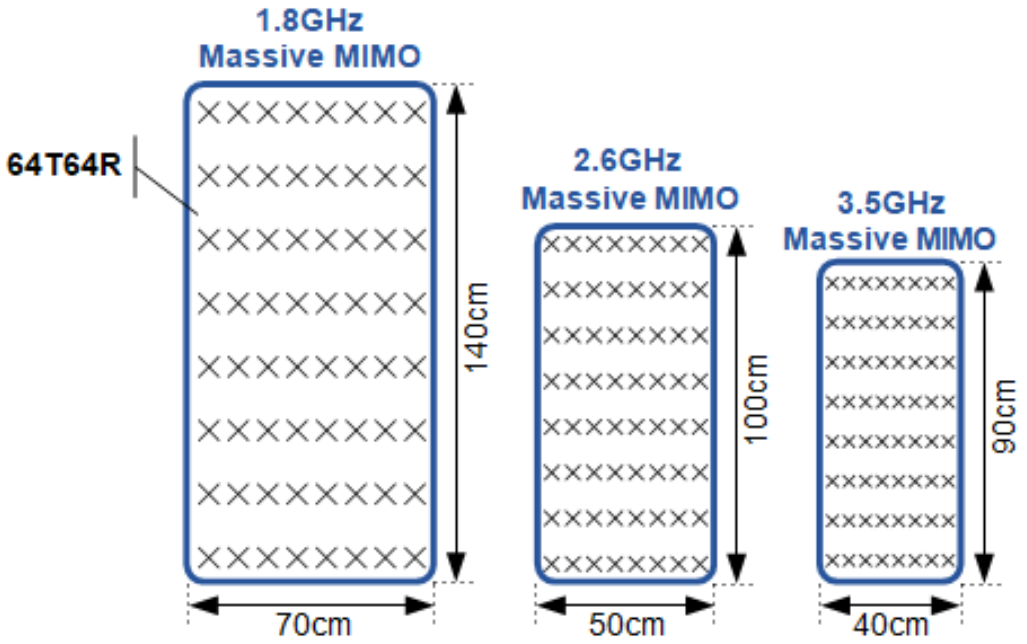


Picture besides illustrates the basic concept of how the electromagnetic waves are reinforced in the main beam direction. This is typically termed the main lobe. It is also worth noting that the spacing between the antenna elements is expressed in wavelengths (λ), with different spacing having a different result.



4.5 Below 6GHz Massive MIMO Antenna

Massive MIMO is mostly linked to the requirement to have beamforming at the mmWave frequencies. However, the concept can still be used in the sub- 6GHz band. Since antenna element size and spacing is a factor of wavelength, the size of the antenna will increase as the frequency gets lower (assuming the same level of performance and control is required). Figure besides illustrates typical Massive MIMO antenna sizes for sub 6GHz operation. Each antenna has 64T64R, meaning 64 RF Transceiver Channels.



The key benefit of using 64T64R over traditional 8T8R is an increase to the peak per-carrier capacity, an increase in cell-edge capacity and an extension in overall coverage.

4.6 Above 6GHz Massive MIMO Antenna

In frequencies above 6GHz, as well as mmWave (above 20GHz), the solutions focus on TDD operation with up to 512 ports.



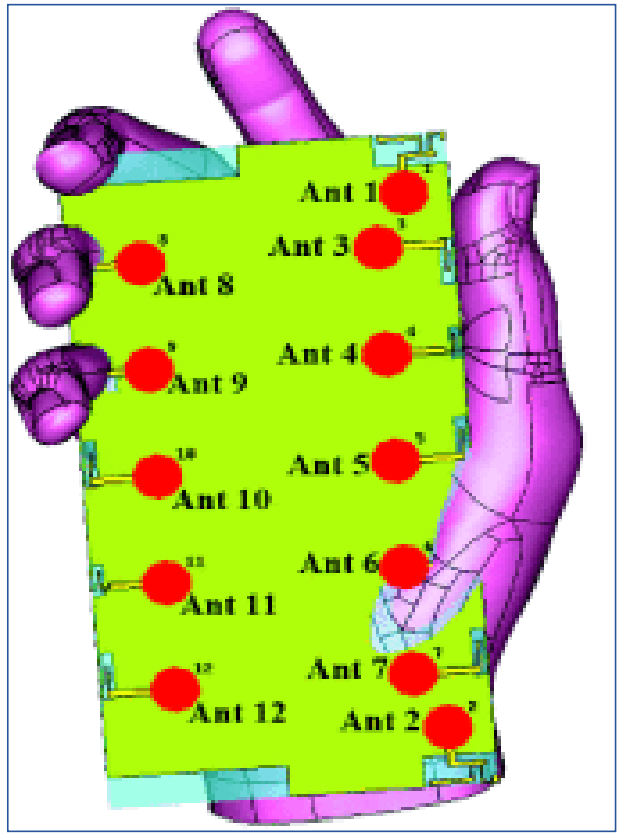
Massive MIMO 32, 64 or 128 Ports

Massive MIMO 256 Ports

Massive MIMO with 512 Ports

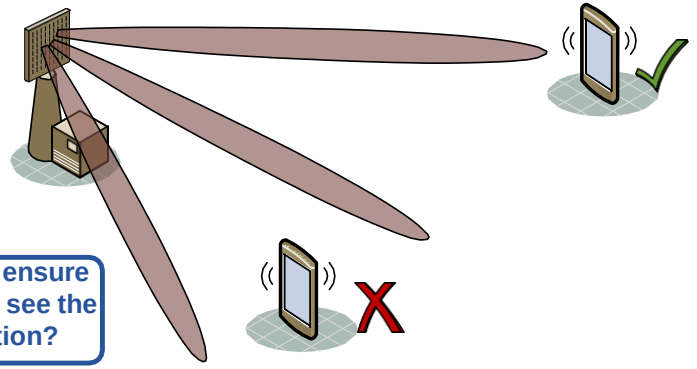
4.7 MIMO Antenna In Handset

5G devices, such as handheld mobiles are also being targeted for MIMO enhancements, as well as the use of beam manipulation. Figure besides illustrates a scenario with multiple antenna elements on a handheld device. These could then be used to manipulate the radiation pattern around the hand, as well as providing limited steering capability towards the cell.



4.8 Beam Management in 5G NR

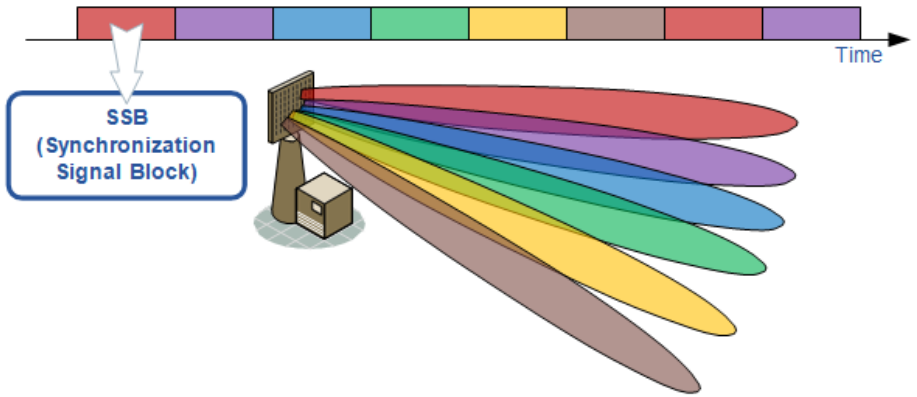
1



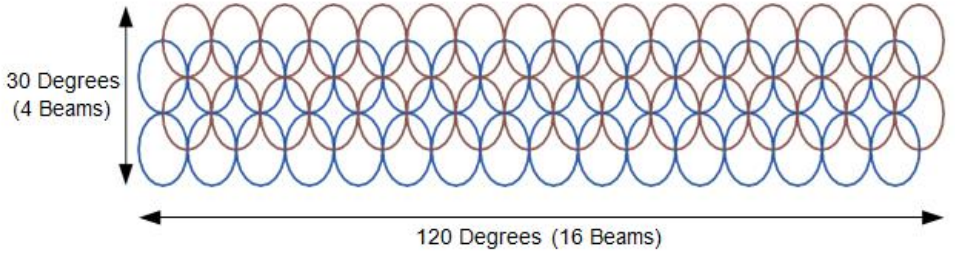
2

Beam Sweeping	The process of covering a spatial area utilizing a set of beams which are transmitted/received at pre-specified intervals and directions.
Beam Measurement	The evaluation of the quality of the received signal. Either at the gNB or at the device.
Beam Determination	The selection of the suitable beam and is related to the measurements obtained with beam measurement.
Beam Reporting	The procedure used by the device to send beam quality and beam decision information.

3



4

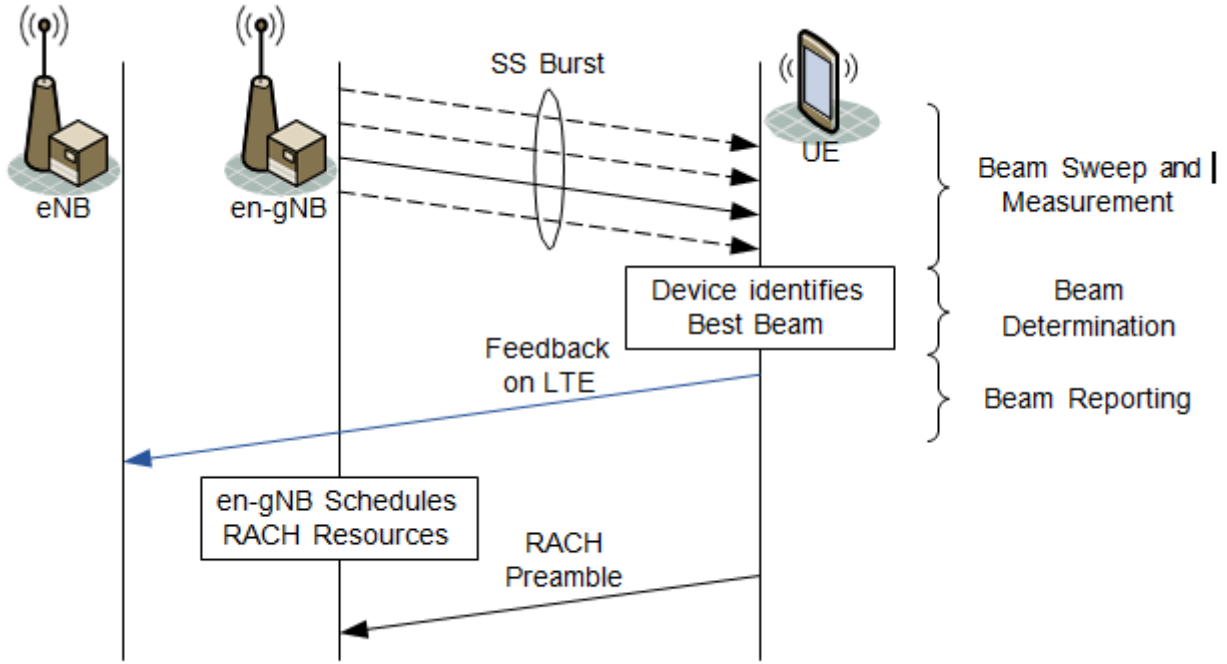


The 5G NR system defines a SSB (Synchronization Signal Block). This includes NR synchronization signals and a PBCH (Physical Broadcast Channel).

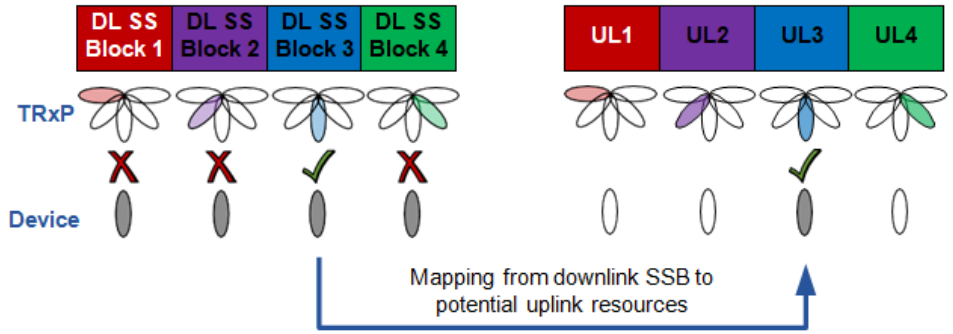
Due to the narrow beams, it is important that the beam sweep covers both the azimuth of the cell's footprint, as well as the vertical footprint.

4.9 Beam Management in 5G NR

5



6

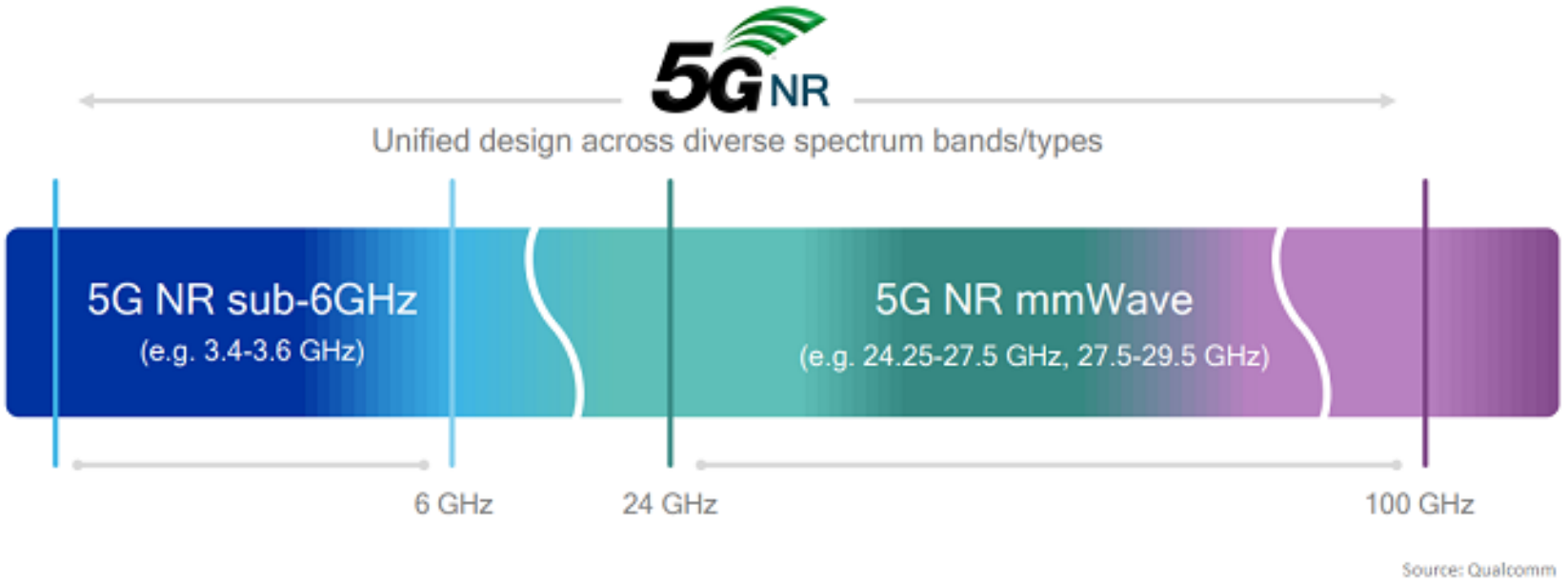


During the initial discovery procedure, beam measurement and beam determination is based on the SS/PBCH Block. For EN-DC operation, since the device does not establish the RRC connection towards the en-gNB, the initial beam reporting process will go via the LTE master node. This in turn can communicate with the en-gNB to schedule RACH resources.

5. Frequency Band

In 5G Frequency Band Requirements throughout the RF specifications are in many cases defined separately for different **Frequency Ranges (FR)** like table below :

Frequency range designation	Corresponding frequency range	Ghz
FR1	450 MHz – 6000 MHz	0.4 Ghz-6 Ghz
FR2	24250 MHz – 52600 MHz	24.25 Ghz to 52.6 Ghz



5.1 Operating Bands in FR1

NR Operating Band	Uplink (UL) operating band BS receive UE transmit		Downlink (DL) operating band BS transmit UE receive		Duplex Mode
	F_{UL_low} – F_{UL_high}	total BW	F_{DL_low} – F_{DL_high}	total BW	
N1	1920 MHz-1980 MHz	60	2110 MHz-2170 MHz	60	FDD
N2	1850 MHz-1910 MHz	60	1930 MHz-1990 MHz	60	FDD
N3	1710 Mhz -1785 MHz	75	1805 MHz-1880 MHz	75	FDD
N5	824 -849 MHz	25	869 MHz-894MHz	25	FDD
N7	2500 MHz-2570 MHz	70	2620 MHz-2690 MHz	70	FDD
N8	880 MHz-915 MHz	35	925 MHz-960 MHz	35	FDD
N20	832 MHz-862 MHz	30	791 MHz-821 MHz	30	FDD
N28	703 MHz-748 MHz	45	758 MHz-803 MHz	45	FDD
N38	2570 MHz-2620 MHz	50	2570 MHz-2620 MHz	50	TDD
N41	2496 MHz-2690 MHz	194	2496 MHz-2690 MHz	194	TDD
N50	1432 MHz-1517 MHz	85	1432 MHz-1517 MHz	85	TDD
N51	1427 MHz-1432 MHz	5	1427 MHz-1432 MHz	5	TDD
N66	1710 MHz-1780 MHz	70	2110 MHz-2200 MHz	90	FDD
N70	1695 MHz-1710 MHz	15	1995 MHz-2020 MHz	25	FDD
N71	663 MHz-698 MHz	35	617 MHz-652 MHz	35	FDD
N74	1427 MHz-1470 MHz	43	1475 MHz-1518 MHz	43	FDD

NR Operating Band	Uplink (UL) operating band BS receive UE transmit		Downlink (DL) operating band BS transmit UE receive		Duplex Mode
	F_{UL_low} – F_{UL_high}	total BW	F_{DL_low} – F_{DL_high}	total BW	
N75	N/A		1432 Mhz -1517 MHz	85	SDL
N76	N/A		1427 Mhz -1432 MHz	5	SDL
N78	3300 MHz-3800 MHz	500	3300 MHz - 3800 MHz	500	TDD
N77	3300 MHz- 4200 MHz	900	3300 MHz - 4200 MHz	900	TDD
N79	4400 MHz-5000 MHz	600	4400 MHz - 5000 MHz	600	TDD
N80	1710 MHz-1785 MHz	75	N/A		SUL
N81	880 MHz-915 MHz	35	N/A		SUL
N82	832 MHz-862 MHz	30	N/A		SUL
N83	703 MHz-748 MHz	45	N/A		SUL
N84	1920 MHz-1980 MHz	60	N/A		SUL
N86	1710 MHz-1780 MHz	70	N/A		SUL

FDD = Frequency Division Duplex
TDD = Time Division Duplex
SDL = Supplementary Downlink Bands
SUL = Supplementary Uplink Bands

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5.2 Operating Bands in FR2

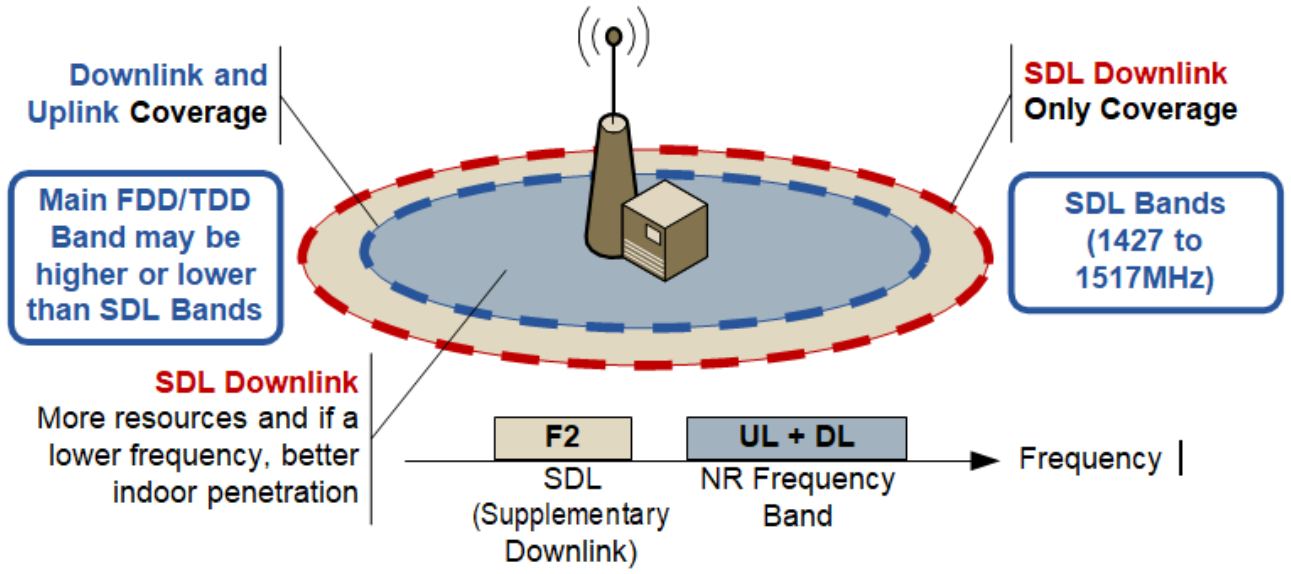
NR Operating Band	Uplink (UL) operating band		Downlink (DL) operating band		Duplex Mode
	BS receive		BS transmit		
	UE transmit		UE receive		
	F_{UL_low} – F_{UL_high}	total BW	F_{DL_low} – F_{DL_high}	total BW	
n257	26500 MHz–29500 MHz	3000	26500 MHz–29500 MHz	3000	TDD
n258	24250 MHz–27500 MHz	3260	24250 MHz–27500 MHz	3260	TDD
n260	37000 MHz–40000 MHz	3000	37000 MHz–40000 MHz	3000	TDD

FDD = Frequency Division Duplex
TDD = Time Division Duplex
SDL = Supplementary Downlink Bands
SUL = Supplementary Uplink Bands

5.3 Supplementary Downlink

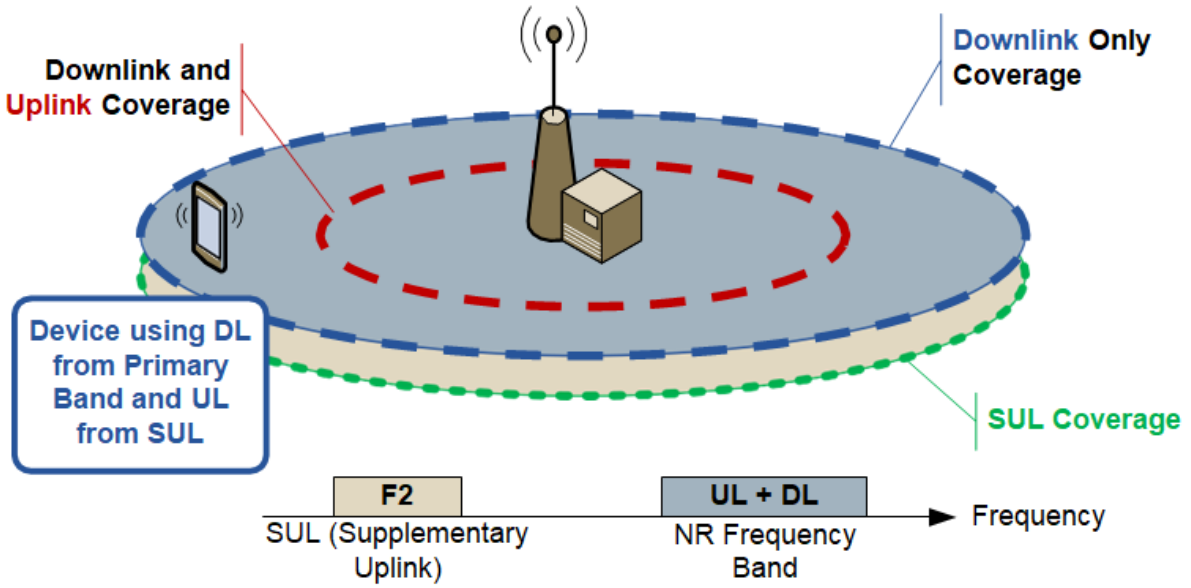
The use of SDL (Supplementary Downlink) is typically related to providing a capacity boost in the downlink. In WRC-15 (World Radio Conference 15), the frequency range 1427MHz to 1517MHz was made available for SDL operation.

Devices inside the blue ring are able to benefit from the SDL operation with greater resource allocation, as well as better indoor penetration (if a lower frequency).

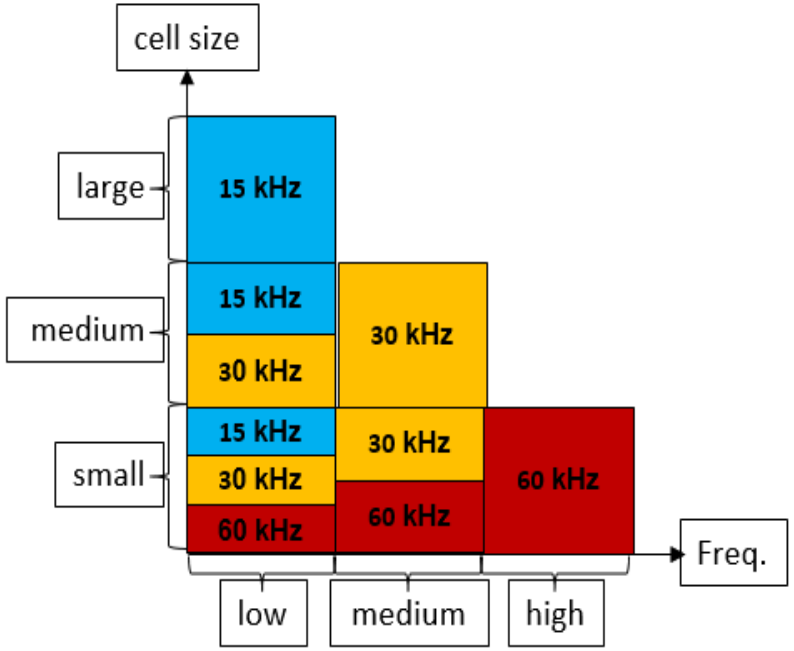


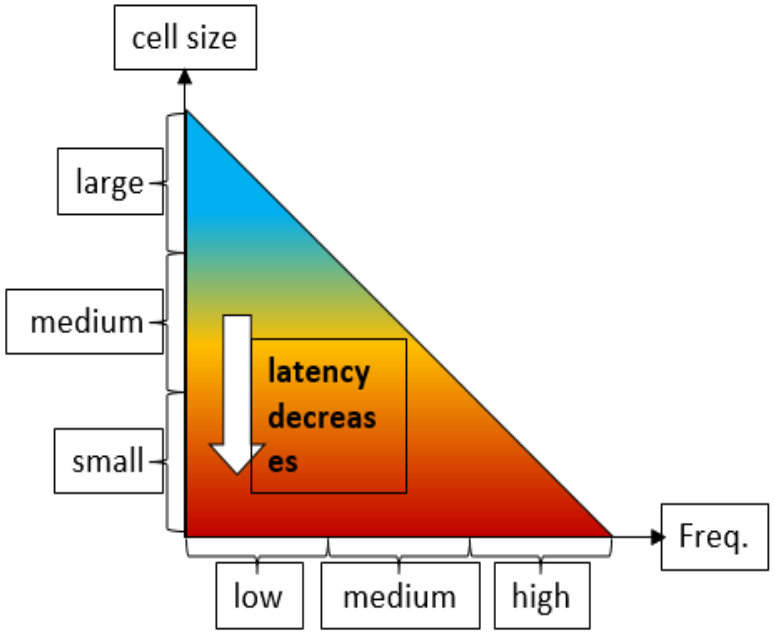
5.4 Supplementary Uplink

The primary band (shaded blue) provides good downlink and uplink coverage, however as a result of device power limitations the uplink coverage is actually reduced (red area). The SUL is typically implemented at a lower frequency band and thus the uplink footprint is much greater (green area). In so doing, devices on the edge of the blue shaded area can utilize the downlink from the primary band and the uplink from the SUL.



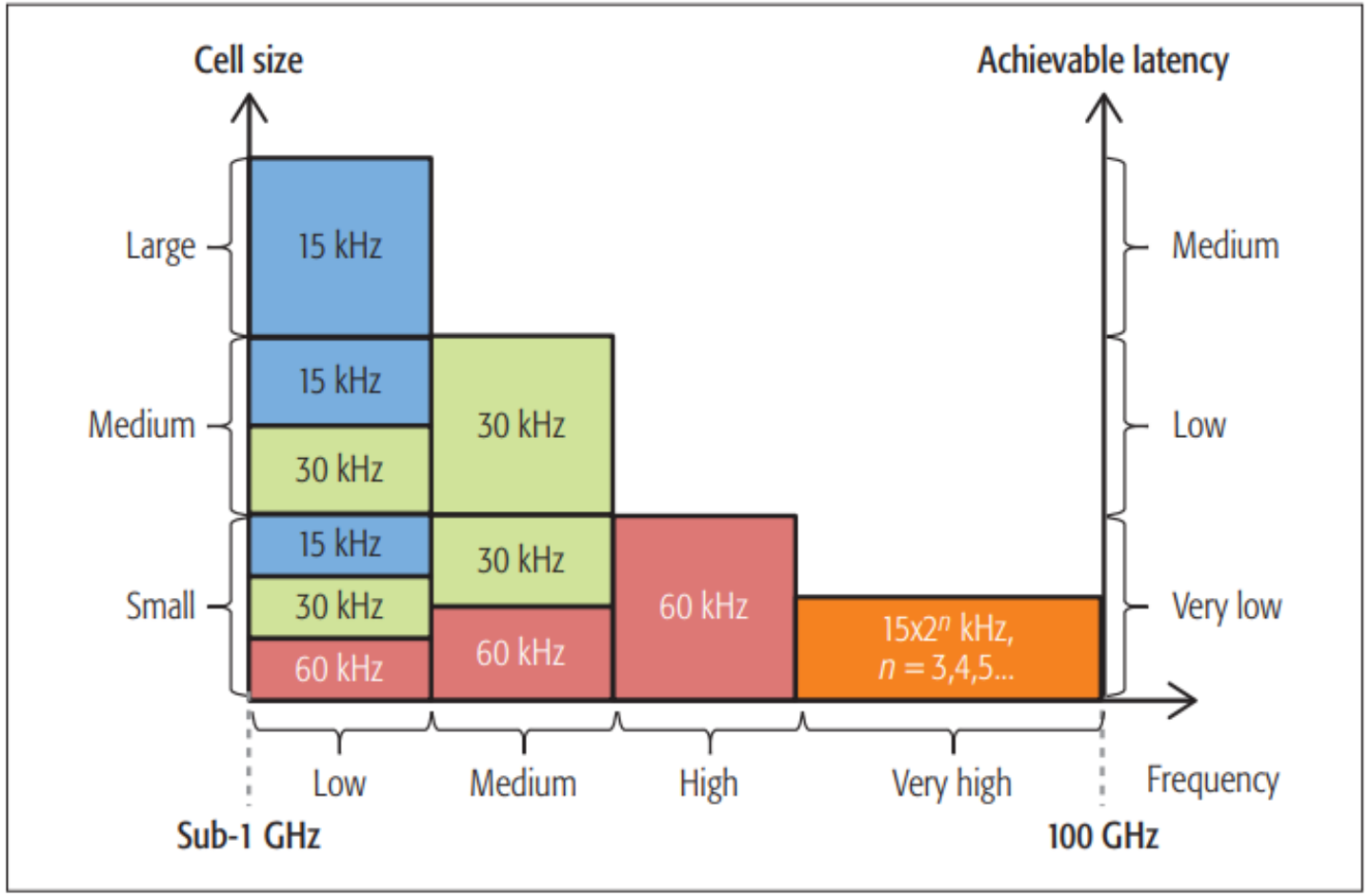
5.5 Numerology Application in NR Band





Very low delays can be realized in small cells using wide subcarriers (in large cells extended CP needed)

5.5 Numerology Application in NR Band



FLOATWAY SYSTEMS

5.6 Channel Bandwidths for Each NR band : FR1

NR band / SCS / UE Channel bandwidth												
NR Band	SCS kHz	5 MHz	10 MHz	15 MHz	20 MHz	25 MHz	30 MHz	40 MHz	50 MHz	60 MHz	80 MHz	100 MHz
N1	15	Yes	Yes	Yes	Yes							
	30		Yes	Yes	Yes							
	60		Yes	Yes	Yes							
N2	15	Yes	Yes	Yes	Yes							
	30		Yes	Yes	Yes							
	60		Yes	Yes	Yes							
N3	15	Yes	Yes	Yes	Yes	Yes	Yes					
	30		Yes	Yes	Yes	Yes	Yes					
	60		Yes	Yes	Yes	Yes	Yes					
N5	15	Yes	Yes	Yes	Yes							
	30		Yes	Yes	Yes							
	60											
N7	15	Yes	Yes	Yes	Yes							
	30		Yes	Yes	Yes							
	60		Yes	Yes	Yes							
N8	15	Yes	Yes	Yes	Yes							
	30		Yes	Yes	Yes							
	60											
N20	15	Yes	Yes	Yes	Yes							
	30		Yes	Yes	Yes							
	60											
N28	15	Yes	Yes	Yes	Yes							
	30		Yes	Yes	Yes							
	60											

NR band / SCS / UE Channel bandwidth												
NR Band	SCS kHz	5 MHz	10 MHz	15 MHz	20 MHz	25 MHz	30 MHz	40 MHz	50 MHz	60 MHz	80 MHz	100 MHz
N38	15	Yes	Yes	Yes	Yes							
	30		Yes	Yes	Yes							
	60		Yes	Yes	Yes							
N41	15		Yes	Yes	Yes			Yes	Yes			
	30		Yes	Yes	Yes			Yes	Yes	Yes	Yes	Yes
	60		Yes	Yes	Yes			Yes	Yes	Yes	Yes	Yes
N50	15	Yes	Yes	Yes	Yes			Yes	Yes			
	30		Yes	Yes	Yes			Yes	Yes	Yes	Yes	
	60		Yes	Yes	Yes							
N51	15	Yes										
	30											
	60											
N66	15	Yes	Yes	Yes	Yes							
	30		Yes	Yes	Yes			Yes				
	60		Yes	Yes	Yes			Yes				
N70	15	Yes	Yes	Yes	Yes	Yes						
	30		Yes	Yes	Yes	Yes						
	60		Yes	Yes	Yes	Yes						
N71	15	Yes	Yes	Yes	Yes							
	30		Yes	Yes	Yes							
	60											

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5.6 Channel Bandwidths for Each NR band : FR1

NR band / SCS / UE Channel bandwidth												
NR Band	SCS kHz	5 MHz	10 MHz	15 MHz	20 MHz	25 MHz	30 MHz	40 MHz	50 MHz	60 MHz	80 MHz	100 MHz
n74	15	Yes	Yes	Yes	Yes							
	30		Yes	Yes	Yes							
	60		Yes	Yes	Yes							
n75	15	Yes	Yes	Yes	Yes							
	30		Yes	Yes	Yes							
	60		Yes	Yes	Yes							
n76	15	Yes										
	30											
	60											
n77	15		Yes		Yes			Yes	Yes			
	30		Yes		Yes			Yes	Yes	Yes	Yes	Yes
	60		Yes		Yes			Yes	Yes	Yes	Yes	Yes
n78	15		Yes		Yes			Yes	Yes			
	30		Yes		Yes			Yes	Yes	Yes	Yes	Yes
	60		Yes		Yes			Yes	Yes	Yes	Yes	Yes
n79	15							Yes	Yes			
	30							Yes	Yes	Yes	Yes	Yes
	60							Yes	Yes	Yes	Yes	Yes
n80	15	Yes	Yes	Yes	Yes	Yes	Yes					
	30		Yes	Yes	Yes	Yes	Yes					
	60		Yes	Yes	Yes	Yes	Yes					
n81	15	Yes	Yes	Yes	Yes							
	30		Yes	Yes	Yes							
	60											

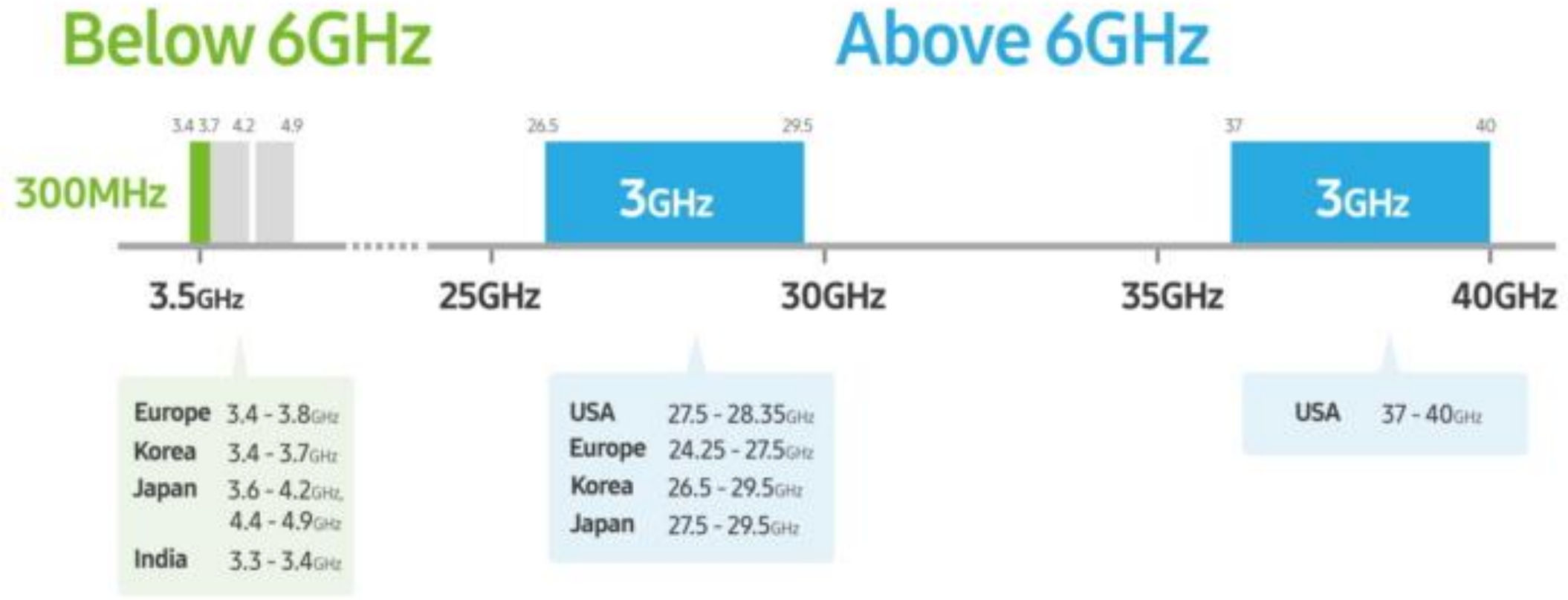
NR band / SCS / UE Channel bandwidth												
NR Band	SCS kHz	5 MHz	10 MHz	15 MHz	20 MHz	25 MHz	30 MHz	40 MHz	50 MHz	60 MHz	80 MHz	100 MHz
n82	15	Yes	Yes	Yes	Yes							
	30		Yes	Yes	Yes							
	60											
n83	15	Yes	Yes	Yes	Yes							
	30		Yes	Yes	Yes							
	60											
n84	15	Yes	Yes	Yes	Yes							
	30		Yes	Yes	Yes							
	60		Yes	Yes	Yes							

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5.7 Channel Bandwidths for Each NR band : FR2

NR band / SCS / UE Channel bandwidth					
NR Band	SCS (kHz)	50 MHz	100 MHz	200 MHz	400 MHz
n257	60	Yes	Yes	Yes	Yes
	120	Yes	Yes	Yes	Yes
n258	60	Yes	Yes	Yes	Yes
	120	Yes	Yes	Yes	Yes
n260	60	Yes	Yes	Yes	Yes
	120	Yes	Yes	Yes	Yes

5.8 Country Wise Spectrum



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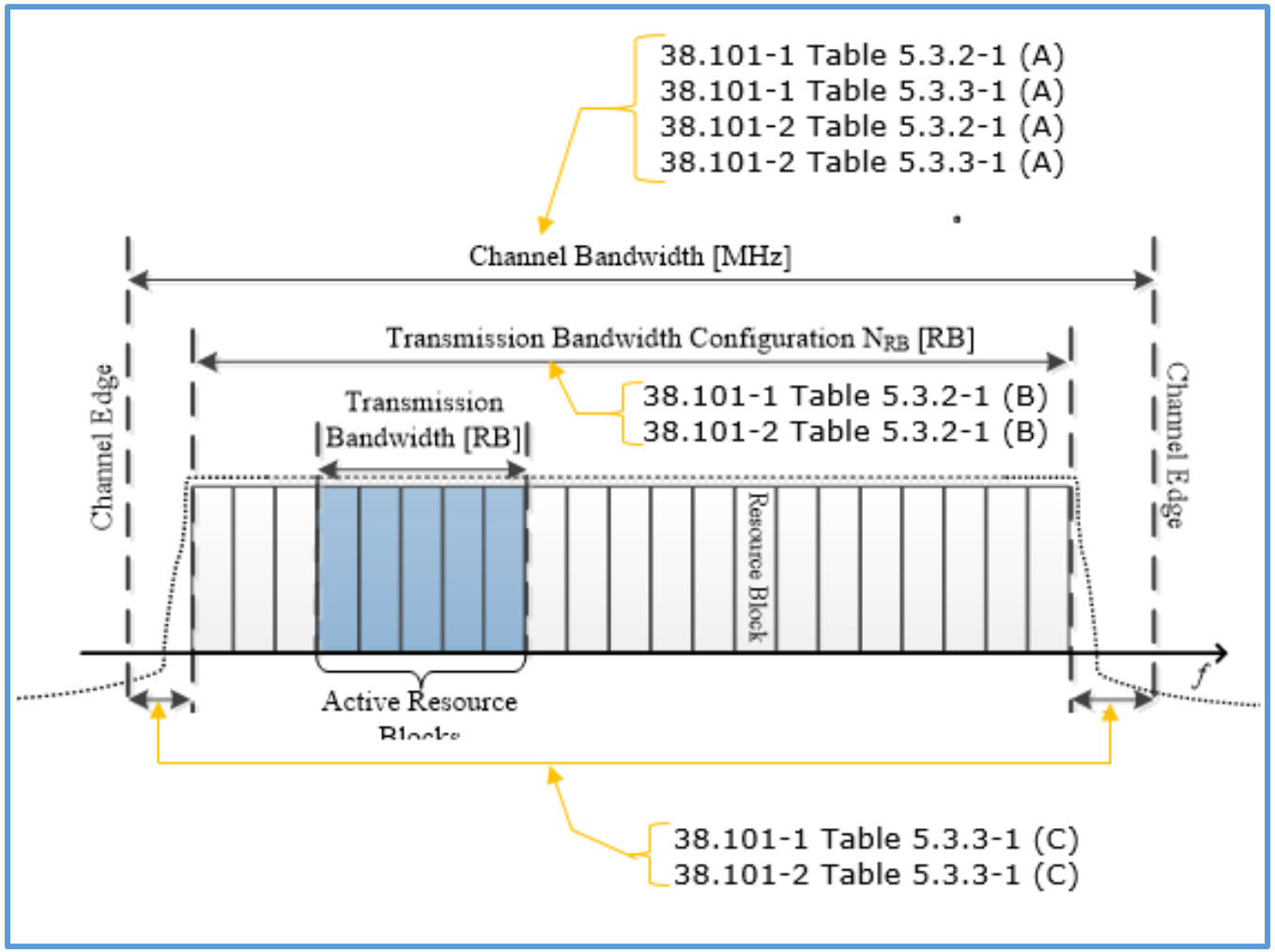
5.8 Country Wise Spectrum

Country / Region	Sub-6 GHz Spectrum (Low and mid band)	Spectrum Above 6 GHz (High band)
USA	600MHz 3100 – 3550 MHz 3700 – 4200 MHz	27.5 – 28.35 GHz 37 – 40 GHZ 64 – 71 GHz
UK / Europe	700 MHz (UK) 2.3 GHz (UK) 3.4 GHz (UK) 694 – 790 MHz 3400 – 3800 MHz	24.25 – 27.5 GHz
Japan	3600 – 4200 MHz 4400 – 4900 MHz	27.5 – 28.25 GHz
China	3300 – 3600 MHz 4400 – 4500 MHz 4800 – 4990 MHz	24.25 – 27.5 GHz 37.25 – 43.5 GHz
Korea	3420MHz-3700MHz	26.5 – 29.5 GHz
India	3400 – 3600 MHz	24.25 - 27.5 GHz
	3300-3400 MHz	37-40.5 GHz
		37-40.5 GHz

5.9 Spectrum and Bandwidth

Channel BW = Maximum transmission bandwidth + guardbands on both side

Channel Bandwidth(CBW) is about [N_RB x NumOfSubcarrier x SCS + GuardBand x 2



5. 10 Maximum Transmission Bandwidth Configuration NRB

Frequency Range 1

(B)

SCS (kHz)	5 MHz	10 MHz	15 MHz	20 MHz	25 MHz	30 MHz	40 MHz	50 MHz	60 MHz	80 MHz	100 MHz
	N_{RB}	N_{RB}	N_{RB}	N_{RB}	N_{RB}	N_{RB}	N_{RB}	N_{RB}	N_{RB}	N_{RB}	N_{RB}
15	25	52	79	106	133	[TBD]	216	270	N/A	N/A	N/A
30	11	24	38	51	65	[TBD]	106	133	162	217	273
60	N/A	11	18	24	31	[TBD]	51	65	79	107	135

(A) (C)

Frequency Range 2

(B)

SCS (kHz)	50MHz	100MHz	200MHz	400 MHz
	N_{RB}	N_{RB}	N_{RB}	N_{RB}
60	66	132	264	N.A
120	32	66	132	264

(A) (C)

5.11 Minimum Guard band for each UE channel bandwidth and SCS (kHz)

Frequency Range 1

(B)

SCS (kHz)	5 MHz	10 MHz	15 MHz	20 MHz	25 MHz	40 MHz	50 MHz	60 MHz	80 MHz	100 MHz
15	242.5	312.5	382.5	452.5	522.5	552.5	692.5	N/A	N/A	N/A
30	505	665	645	805	785	905	1045	825	925	845
60	N/A	1010	990	1330	1310	1610	1570	1530	1450	1370

(A) (C)

Frequency Range 2

(B)

SCS (kHz)	50MHz	100MHz	200MHz	400 MHz
60	1210	2450	4930	N.A
120	1900	2420	4900	9860

(A) (C)

Calculate 5G NR Throughput



5G Throughput Calculation

$$\text{data rate (in Mbps)} = 10^{-6} \cdot \sum_{j=1}^{J} \left(v_{Layers}^{(j)} \cdot Q_m^{(j)} \cdot f^{(j)} \cdot R_{\max} \cdot \frac{N_{PRB}^{BW(j),\mu} \cdot 12}{T_s^{\mu}} \cdot \left(1 - OH^{(j)}\right) \right)$$

1. 10^{-6} This is just a number 10 to power minus 6 and that value is 0.000001

2. $\sum_{j=1}^{J}$ This Value means sum of all carriers & In NR, carrier is the center frequency of the spectrum being used to transmit the NR signal. More spectrum, higher the speed

3. $v_{Layers}^{(j)}$ This means is the maximum number of layers (This means if gNB is transmissiting two antenna to UE, 2X2 or 4X4 , it means how many layers your transmit from gNB to UE)

4. $Q_m^{(j)}$ This value means the maximum modulation order

Modulation scheme
QPSK – 2 bit
16 QAM- 4 bit
64 QAM – 6 bit
256 QAM – 8 bit

5. $f^{(j)}$ This is the scaling factor

The scaling factor can take the values 1, 0.8, 0.75, and 0.4

Note : J is the number of aggregated component carriers in a band or band combination

5G Throughput Calculation

$$\text{data rate (in Mbps)} = 10^{-6} \cdot \sum_{j=1}^{J} \left(v_{Layers}^{(j)} \cdot Q_m^{(j)} \cdot f^{(j)} \cdot R_{\max} \cdot \frac{N_{PRB}^{BW(j),\mu} \cdot 12}{T_s^{\mu}} \cdot \left(1 - OH^{(j)}\right) \right)$$

6. $R_{\max}$ 948/1024 , Coding Rate

7. T_s^{μ} This is the average OFDM symbol duration in a sub frame for numerology

$$T_s^{\mu} = \frac{10^{-3}}{14 \cdot 2^{\mu}}$$ Note that normal cyclic prefix is assumed

μ Numerology 0,1,2,3,4,5

8. $N_{PRB}^{BW(j),\mu} \cdot 12$ This is the maximum RB allocation in bandwic $BW^{(j)}$ BW with numerology μ ,

The maximum transmission bandwidth configuration NRB for each UE channel bandwidth and subcarrier spacing is specified in Table 5.3.2-1.

Maximum transmission bandwidth configuration

SCS (kHz)	5MHz	10MHz	15MHz	20 MHz	25 MHz	30 MHz	40 MHz	50MHz	60 MHz	80 MHz	90 MHz	100 MHz
	N_{RB}	N_{RB}	N_{RB}	N_{RB}	N_{RB}	N_{RB}	N_{RB}	N_{RB}	N_{RB}	N_{RB}	N_{RB}	N_{RB}
15	25	52	79	106	133	160	216	270	N/A	N/A	N/A	N/A
30	11	24	38	51	65	78	106	133	162	217	245	273
60	N/A	11	18	24	31	38	51	65	79	107	121	135

12 This is just number which is multiply by number of PRB

5G Throughput Calculation

$$\text{data rate (in Mbps)} = 10^{-6} \cdot \sum_{j=1}^{J} \left(v_{Layers}^{(j)} \cdot Q_m^{(j)} \cdot f^{(j)} \cdot R_{\max} \cdot \frac{N_{PRB}^{BW(j),\mu} \cdot 12}{T_s^{\mu}} \cdot \left(1 - OH^{(j)}\right) \right)$$

$\left(1 - OH^{(j)}\right)$ Last calculation is 1 minus overhead

1 This is just a number

Table 5.1-1: Definition of frequency ranges

Frequency range designation	Corresponding frequency range
FR1	450 MHz – 6000 MHz
FR2	24250 MHz – 52600 MHz

$OH^{(j)}$

0.14, for frequency range FR1 for DL
0.18, for frequency range FR2 for DL
0.08, for frequency range FR1 for UL
0.10, for frequency range FR2 for UL

Note: Due to higher band in FR2 DL overhead and UL overhead is higher.

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5G Throughput Calculation Example 1

$$\text{data rate (in Mbps)} = 10^{-6} \cdot \sum_{j=1}^{J} \left(v_{Layers}^{(j)} \cdot Q_m^{(j)} \cdot f^{(j)} \cdot R_{\max} \cdot \frac{N_{PRB}^{BW(j),\mu} \cdot 12}{T_s^{\mu}} \cdot \left(1 - OH^{(j)}\right) \right)$$

$\sum_{j=1}^{J}$ **Carriers = 1**

$f^{(j)}$ **Scaling factor = 1**

$v_{Layers}^{(j)}$ **Layers = 2**

$N_{PRB}^{BW(j),\mu} \cdot 12$ **Band = n1 Maximum number of RB**

$Q_m^{(j)}$ **Modulation = 64QAM= 6 bit**

μ **Sub carrier spacing = 30 KHz (Numerology = 1)**

$OH^{(j)}$ **Overhead = 0.14**

NR Data Rate = 10^{-6} * 1 * 2 * 6 * 1 * (948/1024) * (273 * 12) * (14 * 2^{1}) / 10^{-3} * (1 - 0.14) = 876.4 Mbps

5G Throughput Calculation Example 2

SCS (kHz)	5MHz	10MHz	15MHz	20 MHz	25 MHz	30 MHz	40 MHz	50MHz	60 MHz	80 MHz	90 MHz	100 MHz
	N_{RB}	N_{RB}	N_{RB}	N_{RB}	N_{RB}	N_{RB}	N_{RB}	N_{RB}	N_{RB}	N_{RB}	N_{RB}	N_{RB}
15	25	52	79	106	133	160	216	270	N/A	N/A	N/A	N/A
30	11	24	38	51	65	78	106	133	162	217	245	273
60	N/A	11	18	24	31	38	51	65	79	107	121	135

1. Layer : 4
2. Modulation 256 QAM = Modulation order 8
3. Scaling Factor =1
4. Numerology = 0
5. Bandwidth = 50 Mhz
6. Resource Block= 270 from bandwidth and numerology
7. Overhead = 0.14 for DL

$$\text{data rate (in Mbps)} = 10^{-6} \cdot \sum_{j=1}^{J} \left(v_{Layers}^{(j)} \cdot Q_m^{(j)} \cdot f^{(j)} \cdot R_{max} \cdot \frac{N_{PRB}^{BW(j),\mu} \cdot 12}{T_s^{\mu}} \cdot \left(1 - OH^{(j)}\right) \right)$$

0.000001 × 1 × 4 × 8 × 1 × 948/1024 × 160*12 / $\frac{10^{-3}}{14 \cdot 2^{\mu}}$ × 1-0.14 = 1156.65 mbps

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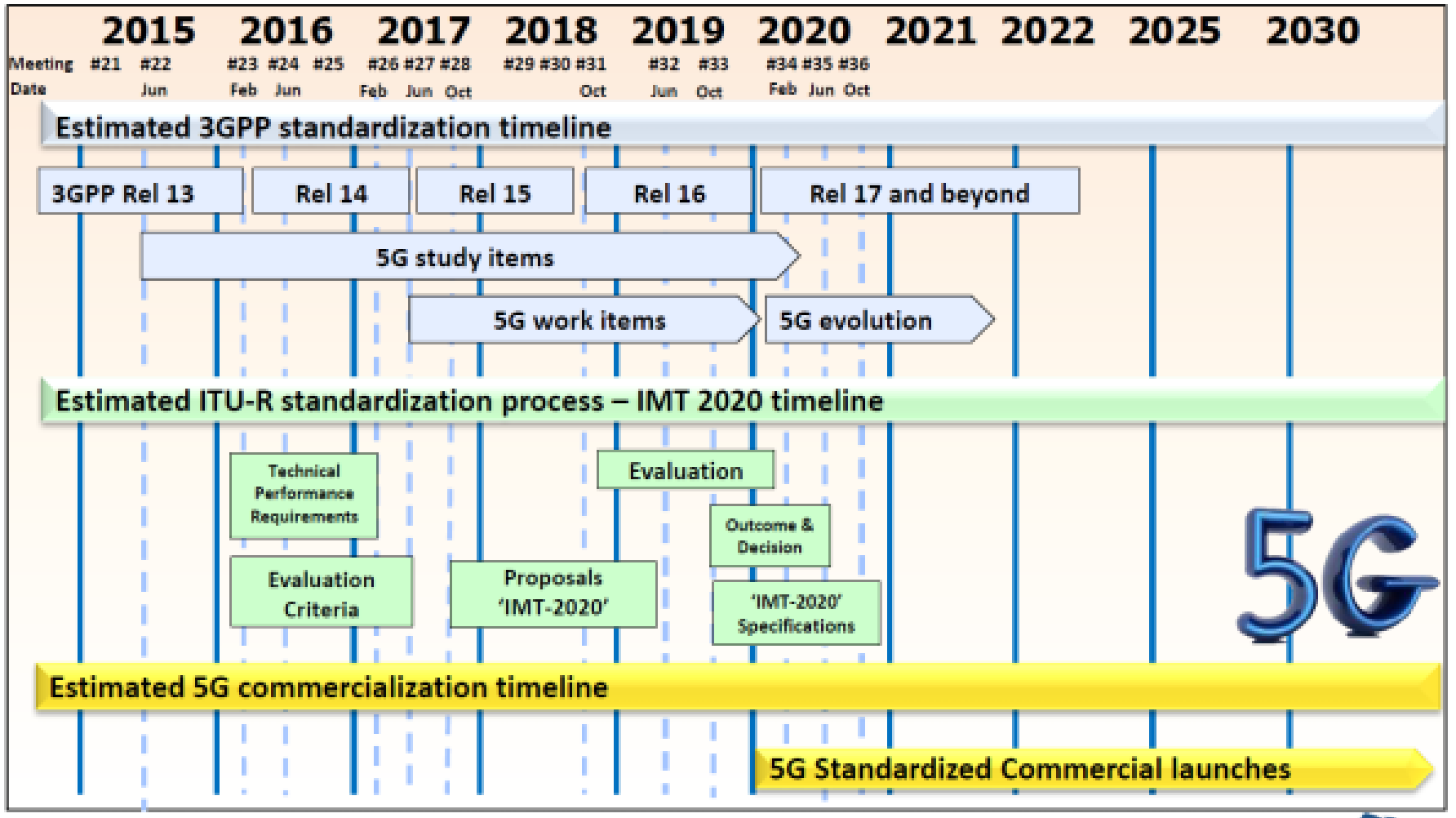
Quiz 5G Throughput Calculation

1. Sebuah operator mengimplementasikan sebuah carrier 5G NR pada N Operating Band N41. Dengan antenna MIMO 64T64R. Hitunglah maksimum throughput UL & DL!
2. Sebuah operator mengimplementasikan dua buah carrier 5G NR pada N Operating Band N77. Dengan antenna MIMO 64T64R. Hitunglah maksimum throughput UL & DL!
3. Sebuah operator mengimplementasikan lima buah carrier 5G NR pada N Operating Band N258. Dengan antenna MIMO 64T64R. Hitunglah maksimum throughput UL & DL!

Roadmap for 5G Standards and Rollout

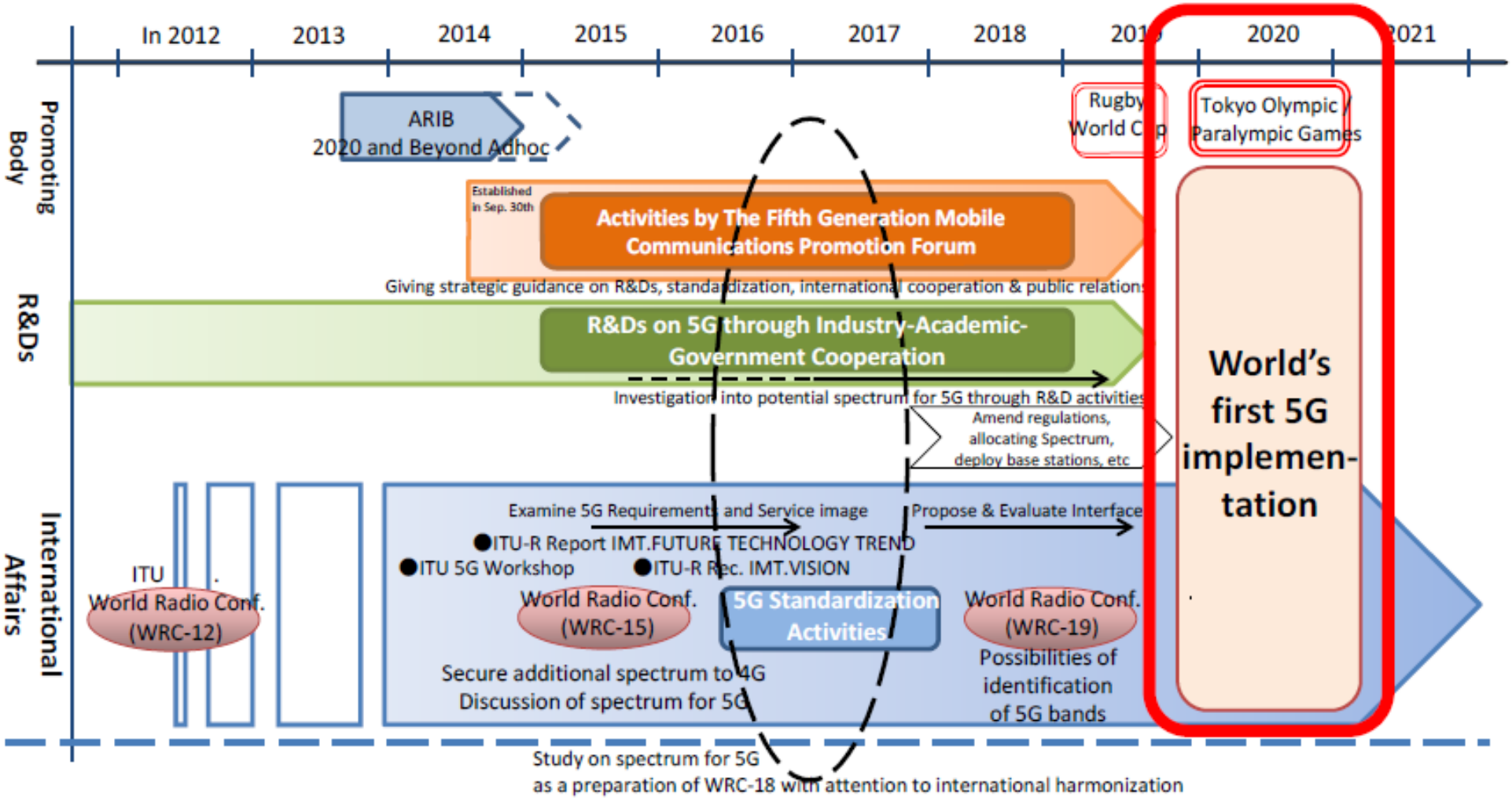


Standardization Timeline



www.4gamericas.org

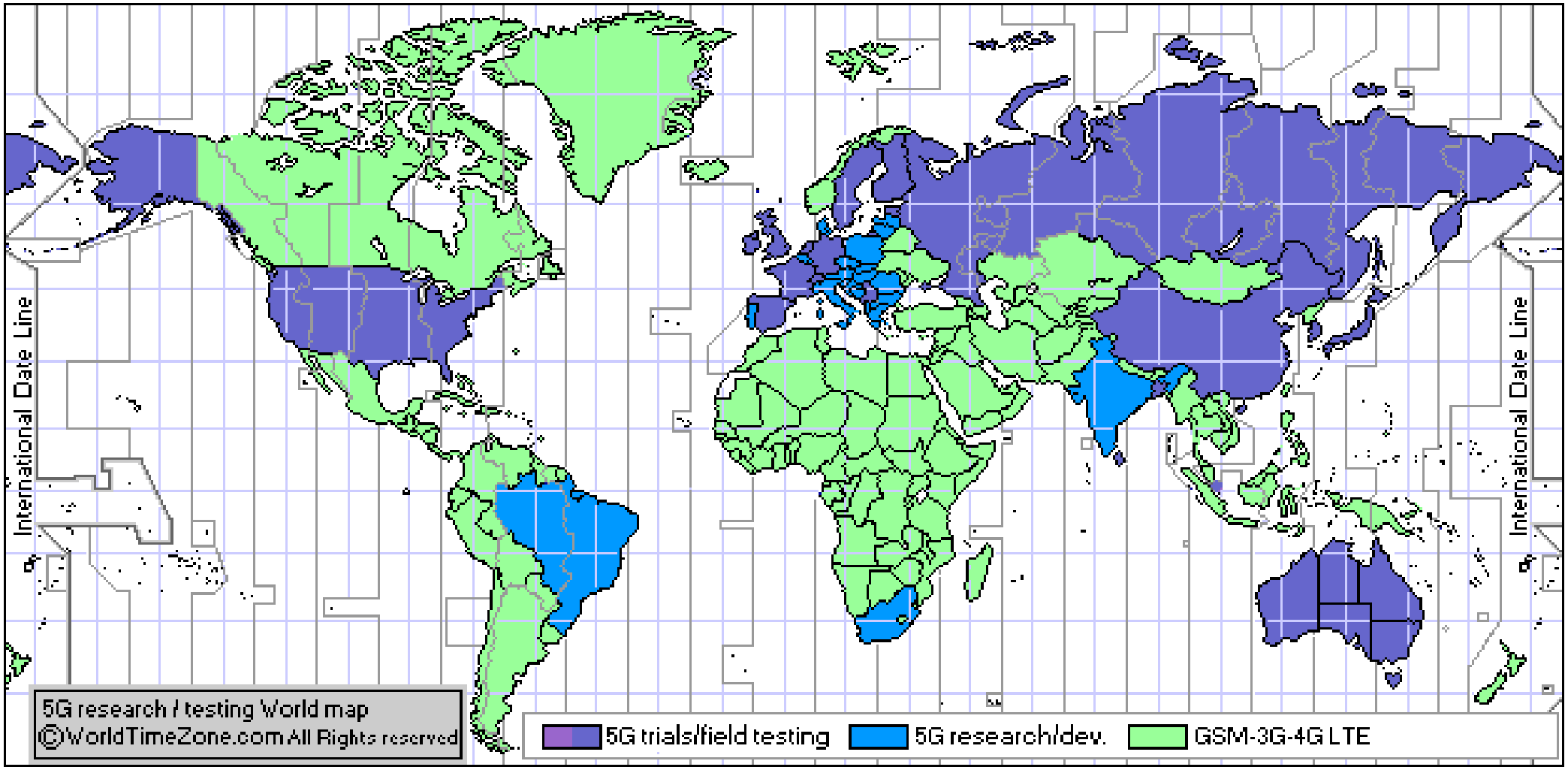
Deployment Timeline : Japan Commercial Launch 2020



Translated from "Final Report from the Radio Policy Vision Council", Ministry of Internal Affairs and Communications, Japan, December 2014.

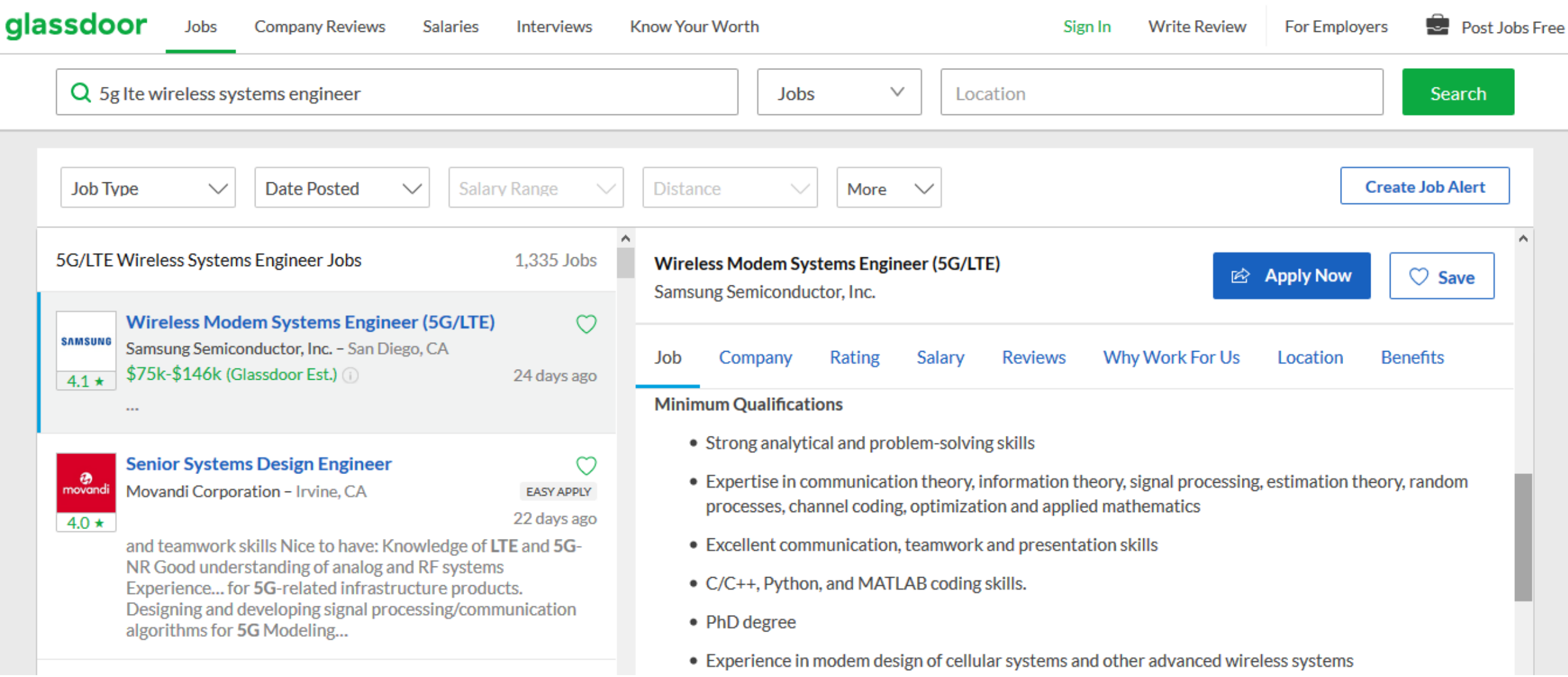
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5G Research /Development → 5G Trial / Field Test → 5G Partial → 5G Commercial → 5G Optimization



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and teamwork skills Nice to have: Knowledge of LTE and 5G-NR Good understanding of analog and RF systems Experience... for 5G-related infrastructure products. Designing and developing signal processing/communication algorithms for 5G Modeling...
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Expertise in communication theory, information theory, signal processing, estimation theory, random processes, channel coding, optimization and applied mathematics
Excellent communication, teamwork and presentation skills
C/C++, Python, and MATLAB coding skills.
PhD degree
Experience in modem design of cellular systems and other advanced wireless systems



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5G Ready Telecom Engineers

Everybody loves speed, right? We already have the advanced network of 4G wireless, but a lot of modern consumers find the speed of this existing network to be sub-par. It is becoming difficult to provide adequate cellular services with ultra-high speed and rapid response due to large file sizes in things like video or music.

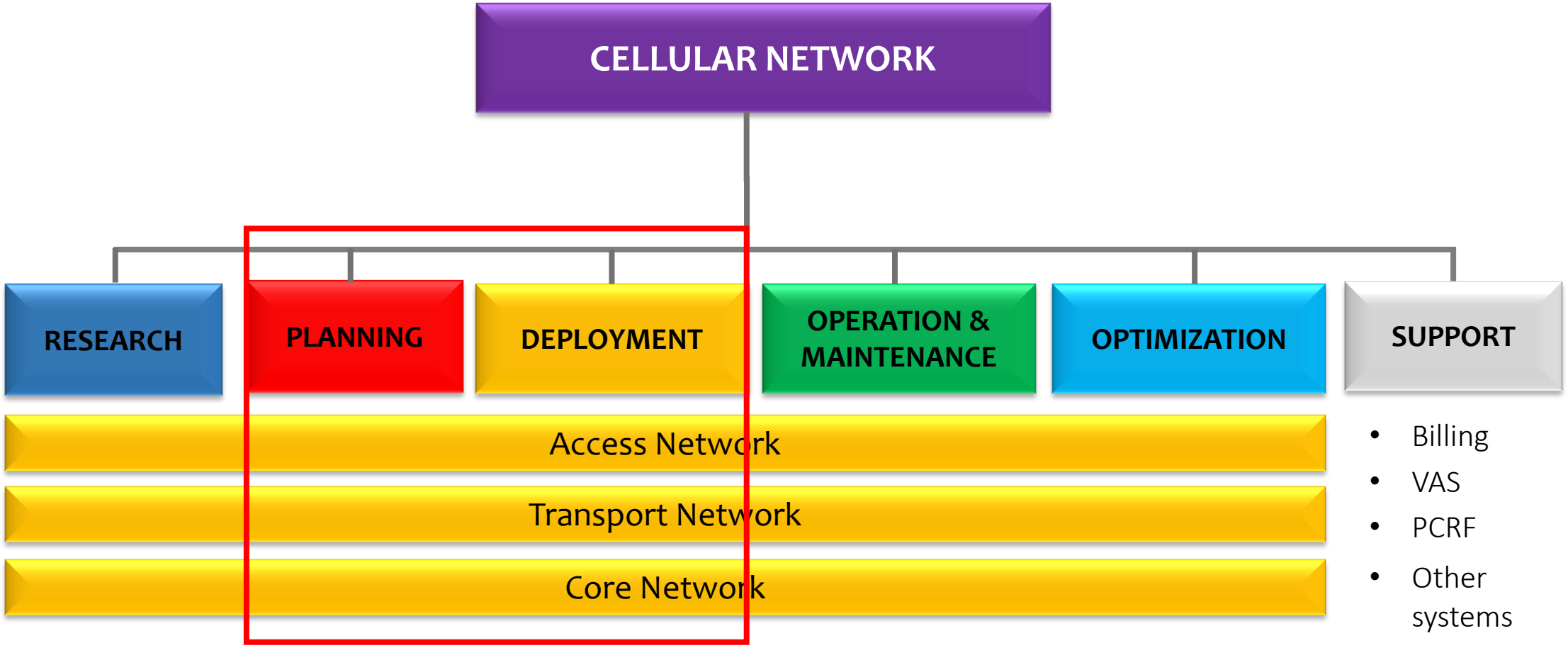
5G also needs to be consistent regarding security and regulation, so engineers will need to stay up to date in this regard. The Engineer will be required to learn and implement new technologies including, but not limited to C-RAN, Massive MIMO, LTE Advanced Pro, SON, HetNet, Mobile Edge Computing, NFV, SDN, Mesh Networks, Network Slicing, and IoT.

Job description

The work of Telecom Engineers has an incredibly large scope and is aimed to improve the existing network. 5G engineers will also be involved in designing and managing the installation of telecom devices and services, like complex electronic switching systems, copper wire telephone cable, fiber optics cabling, or in the implementation of any other internet protocol data system.

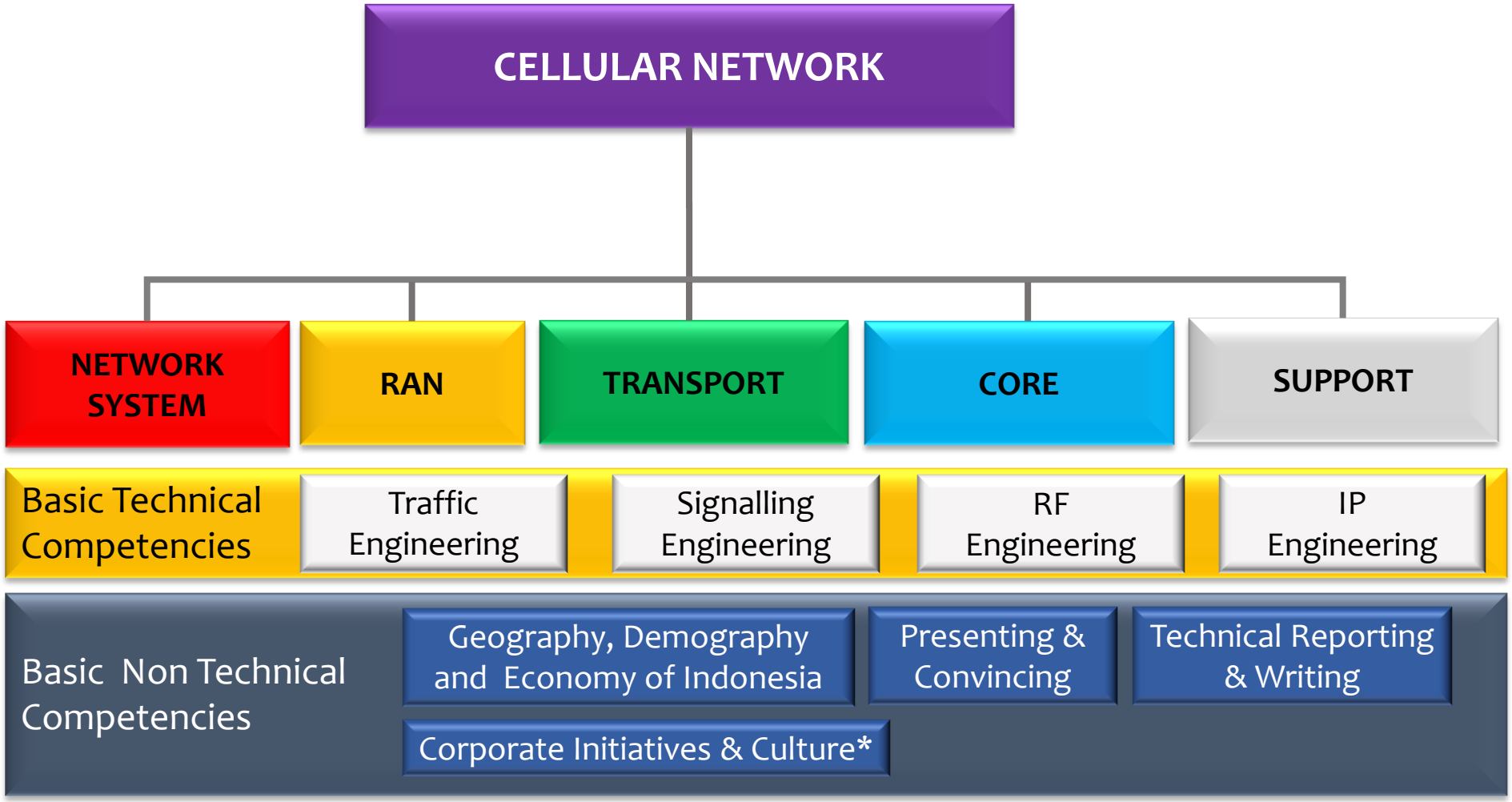
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Process Domain Competencies



Commercial Launch

Network Domain Competencies



*Note:

1.Corporate initiatives should include: Effective Team Interaction, Managing Risk and Efficient Lean Business Process

2.This diagram represent competencies grouping, not organization diagram or scope of accountability

5G Engineer Dreamer (For Global Recruiters)

1. **Network Domain Competencies**
   e.g cmWave, mmWave, OFDM Numerology, 5G Bandwidth, 5G Frequency, New Channel Coding (LDPC and Polar Codes), Flexible Slot, Massive MIMO, NFV (Network Functions Virtualization), SDN (Software Defined Networking), Network Slicing etc
2. **Process Domain Competencies**
   e.g Research, Planning, Deployment, Operation & Maintenance, Optimization, Support
3. **Leadership / Soft Skill**
   e.g Negotiation Skill, Diversity and change management,modern leadership, problem solving and analytical thinking, public speaking/effective communication, creative and critical thinking, decision making, priority and time management, team working
4. **Foreign Language**
   e.g English, Chinese, Arabic
5. **Match with company culture**





Indonesia Use Case





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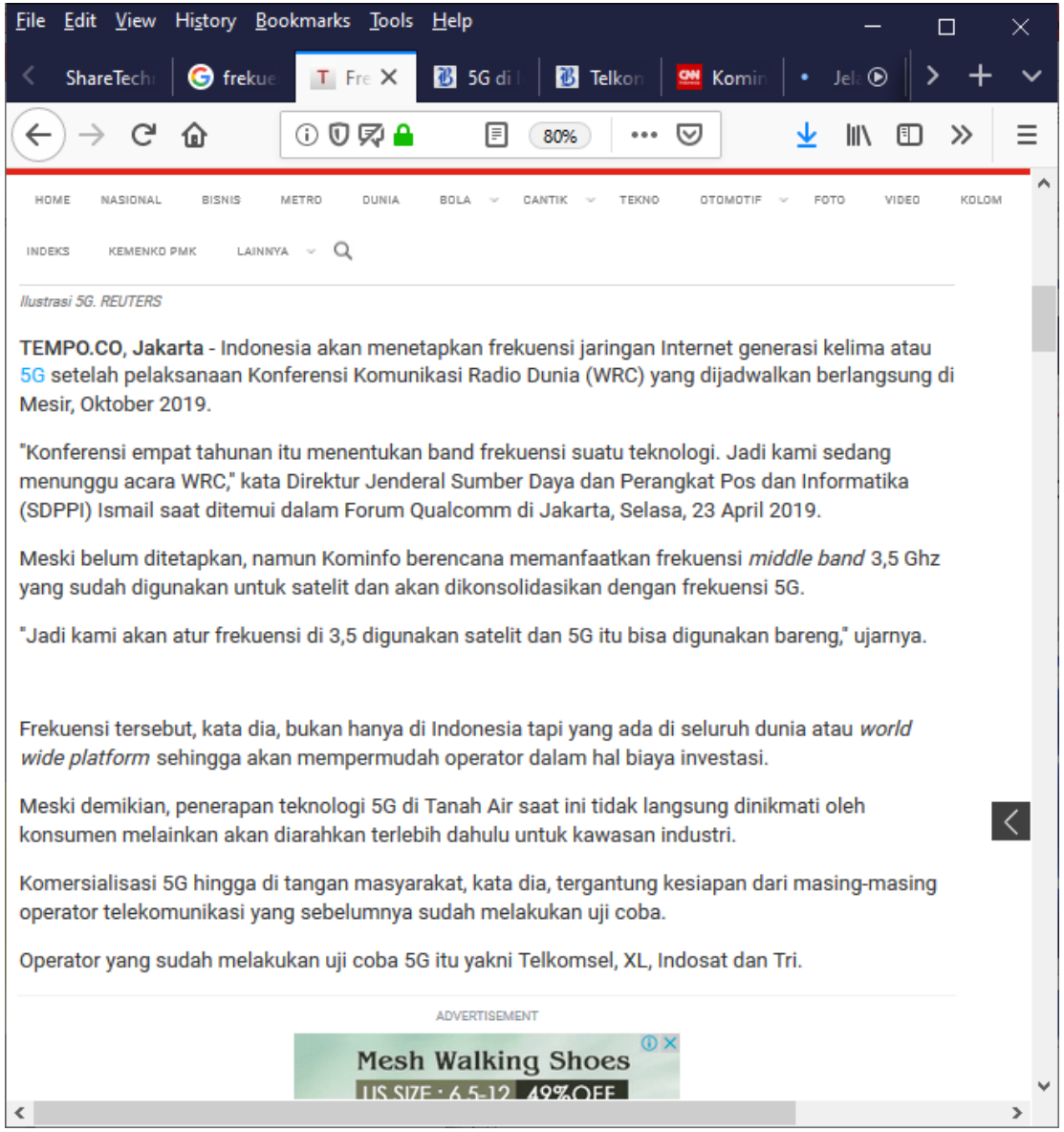
Frekuensi 5G di Indonesia Oktober 2019, 4 Operator Sudah Uji Coba

Reporter: Antara

Editor: Yudono Yanuar

Rabu, 24 April 2019 07:30 WIB

0 KOMENTAR



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INDEKS KEMENKO PMK LAINNYA

Ilustrasi 5G. REUTERS

TEMPO.CO, Jakarta - Indonesia akan menetapkan frekuensi jaringan Internet generasi kelima atau 5G setelah pelaksanaan Konferensi Komunikasi Radio Dunia (WRC) yang dijadwalkan berlangsung di Mesir, Oktober 2019.

"Konferensi empat tahunan itu menentukan band frekuensi suatu teknologi. Jadi kami sedang menunggu acara WRC," kata Direktur Jenderal Sumber Daya dan Perangkat Pos dan Informatika (SDPPI) Ismail saat ditemui dalam Forum Qualcomm di Jakarta, Selasa, 23 April 2019.

Meski belum ditetapkan, namun Kominfo berencana memanfaatkan frekuensi *middle band* 3,5 Ghz yang sudah digunakan untuk satelit dan akan dikonsolidasikan dengan frekuensi 5G.

"Jadi kami akan atur frekuensi di 3,5 digunakan satelit dan 5G itu bisa digunakan bareng," ujarnya.

Frekuensi tersebut, kata dia, bukan hanya di Indonesia tapi yang ada di seluruh dunia atau *world wide platform* sehingga akan mempermudah operator dalam hal biaya investasi.

Meski demikian, penerapan teknologi 5G di Tanah Air saat ini tidak langsung dinikmati oleh konsumen melainkan akan diarahkan terlebih dahulu untuk kawasan industri.

Komersialisasi 5G hingga di tangan masyarakat, kata dia, tergantung kesiapan dari masing-masing operator telekomunikasi yang sebelumnya sudah melakukan uji coba.

Operator yang sudah melakukan uji coba 5G itu yakni Telkomsel, XL, Indosat dan Tri.

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5G di Indonesia: Bisakah Pakai Frekuensi 2,6 GHz?

Frekuensi 3,5GHz yang sempat diwacanakan Kementerian Komunikasi dan Informatika tidak dapat digunakan karena masih dipakai oleh satelit.

Leo Dwi Jatmiko - Bisnis.com
17 Juli 2019 | 08:58 WIB



Bisnis.com, JAKARTA – PT Telekomunikasi Indonesia Tbk. (Telkom) menyarankan kepada pemerintah untuk menggunakan spektrum lain untuk implementasi generasi ke lima atau 5G.

Direktur Utama Telkom Ririek Adriansyah menuturkan bahwa hingga saat ini implementasi 5G di Indonesia masih terhambat oleh alokasi spektrum dan regulasi.

Dia mengatakan bahwa frekuensi 3,5GHz yang sempat diwacanakan Kementerian Komunikasi dan Informatika tidak dapat digunakan karena masih dipakai oleh satelit. Begitupun dengan skema berbagi frekuensi, juga tidak dapat dilakukan karena rawan inteferensi.

Ririek berpendapat selain 3,5 GHz frekuensi yang memungkinkan untuk digunakan 5G adalah frekuensi s-band di 2,6 GHz, dengan melihat jumlah pengguna yang ada saat ini.

“Selain 3.5 mungkin yang paling dekat untuk bisa dipakai adalah 2.6GHz dengan pertimbangan potensi jumlah user secara global di dunia dalam waktu dekat,” kata Ririek kepada *Bisnis.com*, Selasa (16/7/2019).

Diketahui berdasarkan laporan,Ericsson Mobility Report Juni 2019, disebutkan bahwa penyerapan 5G secara global terjadi lebih cepat dari waktu yang diperkirakan sebelumnya pada laporan November 2018, seiring dengan penyebaran 5G oleh operator dan pengguna gawai yang beralih ke 5G.

Laporan tersebut memperkirakan bahwa pelanggan 5G akan mencapai 1,9 miliar pada 2024, angka ini naik 27% dibandingkan dengan perkiraan sebelumnya pada November 2018 yang memprediksi hanya 1,5 miliar pengguna 5G pada 2024.Tidak hanya itu pada 2019 ini, terdapat sejumlah vendor yang telah meluncurkan ponsel pintar yang mendukung jaringan 5G.

Untuk di *high band* pada rentang frekuensi 28 GHz – 39 GHz terdapat dua ponsel pintar yang telah mendukung 5G. Kemudian pada rentang frekuensi mid band frekuensi 3,5 GHz dan 2,6 GHz terdapat dua ponsel juga yang telah mendukung 5G.

Lebih lanjut, berdasarkan data yang diterima dari sumber Bisnis, beberapa operator di China dan Amerika Serikat telah memanfaatkan 2,6 GHz untuk 5G. Sementara itu, Thailand, Vietnam, Myanmar, Norwegia, Arab Saudi, Mesir dan Jepang baru berencana menggunakan 2,6 GHz.

Adapun mengenai implementasi 5G di Indonesia, Ririek menilai kemungkinan baru dapat diterapkan di indutri dengan skema Business to Business (B2B).

Ririek melihat sulit untuk geneasi ke lima dipasarkan secara komersial kepada konsumen mengingat besarnya nilai investasi yang harus digelontorkan untuk mengembangankan jaringan dan harga layanan yang mahal.

Dia menyampaikan bahwa Telkom sudah siap untuk mengimplematasikan 5G dari segi B2B, jika regulasi dan spektrum telah tersedia. Telkom telah menjalin pembicaraan dengan salah perusahaan yang begerak di industri kertas di salah satu pulau mengenai pemanfaatan 5G, dalam bentuk *joint innovation center.*

“Kami siap, meskipun pengembangan jaringan dan layanan 5G ini harus dilakukan secara terukur,” kata Ririek.

Di samping itu, sambungnya, Telkom juga telah berkerja sama dengan beberapa vendor dalam pengembangan pemanfaatan 5G sekaligus uji coba untuk B2B.

Adapun untuk frekuensi yang digunakan untuk B2B dan B2C menurutnya tidak ada perbedaan.

“Kerja sama yang ada dengan beberapa vendor lebih untuk mengembangkan use case [di industri] termasuk trial,” kata Ririek.

KENDALA

Mengenai pemanfaatan frekuensi 2,6 GHz untuk 5G, Ketua Program Studi Magister Teknik Elektro Institut Teknologi Bandung (ITB) Ian Yosef M. Edward menilai bahwa terdapat beberapa tantangan yang harus dihadapi untuk memanfaatkan frekuensi tersebut untuk 5G.

Pertama, Frekuensi 2,6 GHz masih digunakan oleh satelit Indovision SES7 untuk penyiaran. Kedua, belum ada konvergensi antara bisnis penyiaran dengan bisnis telekomunikasi.

“Jika ada konvergensi bisnis keduanya, pengaturannya akan lebih mudah, karena menjadi penyedia kontent, penyedia network dan service provider saja. Saat ini konten provider

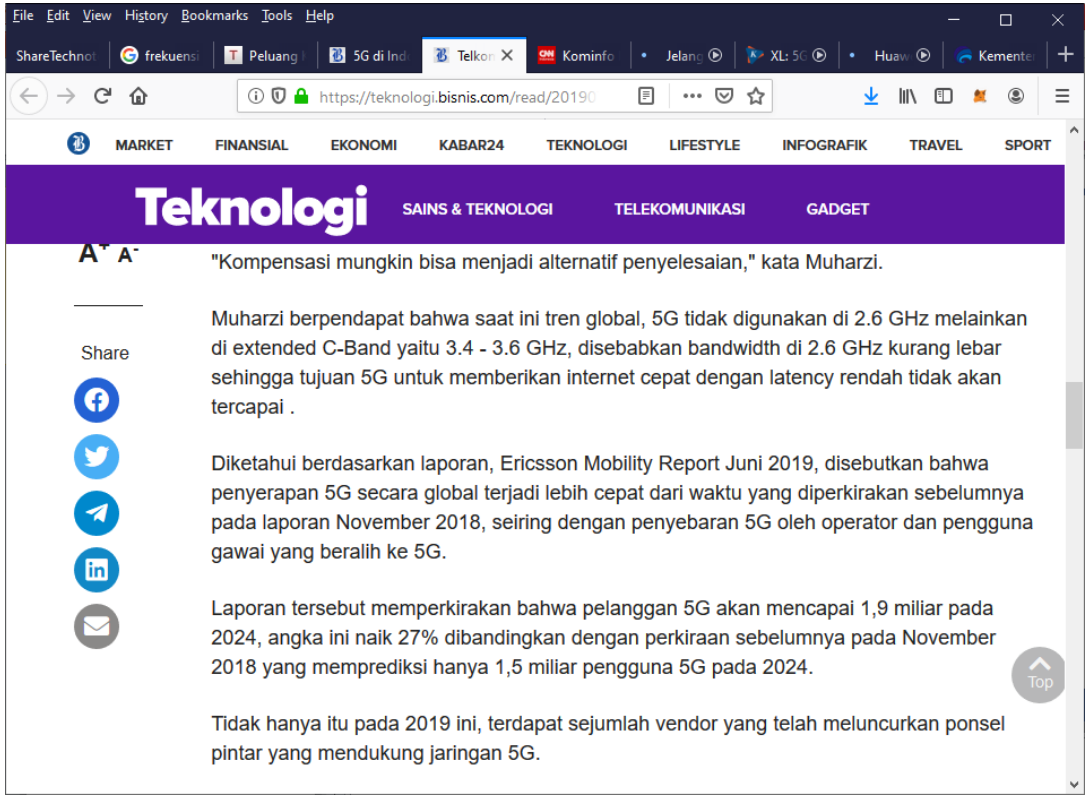
Menteri Komunikasi dan Informatika Rudiantara mengatakan pemerintah bisa saja membuka jaringan 5G dalam waktu dekat lewat *sharing*frekuensi 5G dengan satelit di pita frekuensi 3,5 GHz.

Hanya saja, sambungnya, jaringan 5G yang digelar nantinya hanya mencangkup kawasan-kawasan tertentu yang tidak tercangkup oleh satelit. Adapun jika masyarakat ingin mendapat manfaat sepenuhnya dari frekuensi 3,5Ghz, mereka harus menunggu hingga kontrak frekuensi satelit habis.

“Kita bisa menggunakan 3,5 Ghz tetapi di daerah yang tidak tercangkup oleh satelit Extended C, nah kebanyakan [Extended C] tidak digunakan di Pulau Jawa,” kata Rudiantara.

Salah satu skenario yang dapat dilakukan dalam berbagi frekuensi adalah dengan membagi wilayah operasional frekuensi. 5G yang membutukan kapasitas besar dari serat optik dapat digelar di perkotaan, sedangkan satelit dengan cangkupan yang luas tanpa harus menggelar kabel, seharusnya dapat memberi pelayanan di daerah rural atau pedesaan.


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Teknologi
SAINS & TEKNOLOGI
TELEKOMUNIKASI
GADGET
A+ A-
Share
Sementara itu, PT MNC Sky Vision Tbk, memastikan pemanfaatan teknologi 5G di pita frekuensi 2,6 GHz tidak akan terjadi, selama Satelit Indovision SES7 masih beroperasi.
Muharzi Hasril, Corporate Secretary PT MNC Sky Vision Tbk, mengatakan bahwa pihaknya masih menggunakan frekuensi 2,6 GHz hingga 2028, dengan kemungkinan dapat bertambah lama jika muncul satelit pengganti yang menggunakan frekuensi yang sama.
"Frekuensi 2,6 GHz mungkin saja digunakan oleh 5G selama frekuensi tersebut tidak lagi digunakan oleh Satelit kami di S-Band," kata Muharzi kepada Bisnis.
Muharzi menambahkan pemanfaatan frekuensi dengan skema berbagi atau sharing frekuensi di frekuensi 2,6 GHz juga sulit dilakukan, karena dikhawatirkan terjadi inteferensi atau gangguan dengan frekuensi 5G.
Adapun jika pemerintah atau pengguna 5G ingin memanfaatkan frekuensi tersebut dalam waktu dekat, dia menyarankan agar pihak terkait memberi kompensasi sebagai alternatif.
Top


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"Kompensasi mungkin bisa menjadi alternatif penyelesaian," kata Muharzi.
Muharzi berpendapat bahwa saat ini tren global, 5G tidak digunakan di 2.6 GHz melainkan di extended C-Band yaitu 3.4 - 3.6 GHz, disebabkan bandwidth di 2.6 GHz kurang lebar sehingga tujuan 5G untuk memberikan internet cepat dengan latency rendah tidak akan tercapai .
Diketahui berdasarkan laporan, Ericsson Mobility Report Juni 2019, disebutkan bahwa penyerapan 5G secara global terjadi lebih cepat dari waktu yang diperkirakan sebelumnya pada laporan November 2018, seiring dengan penyebaran 5G oleh operator dan pengguna gawai yang beralih ke 5G.
Laporan tersebut memperkirakan bahwa pelanggan 5G akan mencapai 1,9 miliar pada 2024, angka ini naik 27% dibandingkan dengan perkiraan sebelumnya pada November 2018 yang memprediksi hanya 1,5 miliar pengguna 5G pada 2024.
Tidak hanya itu pada 2019 ini, terdapat sejumlah vendor yang telah meluncurkan ponsel pintar yang mendukung jaringan 5G.
Top

Role Play Indonesia Use Case

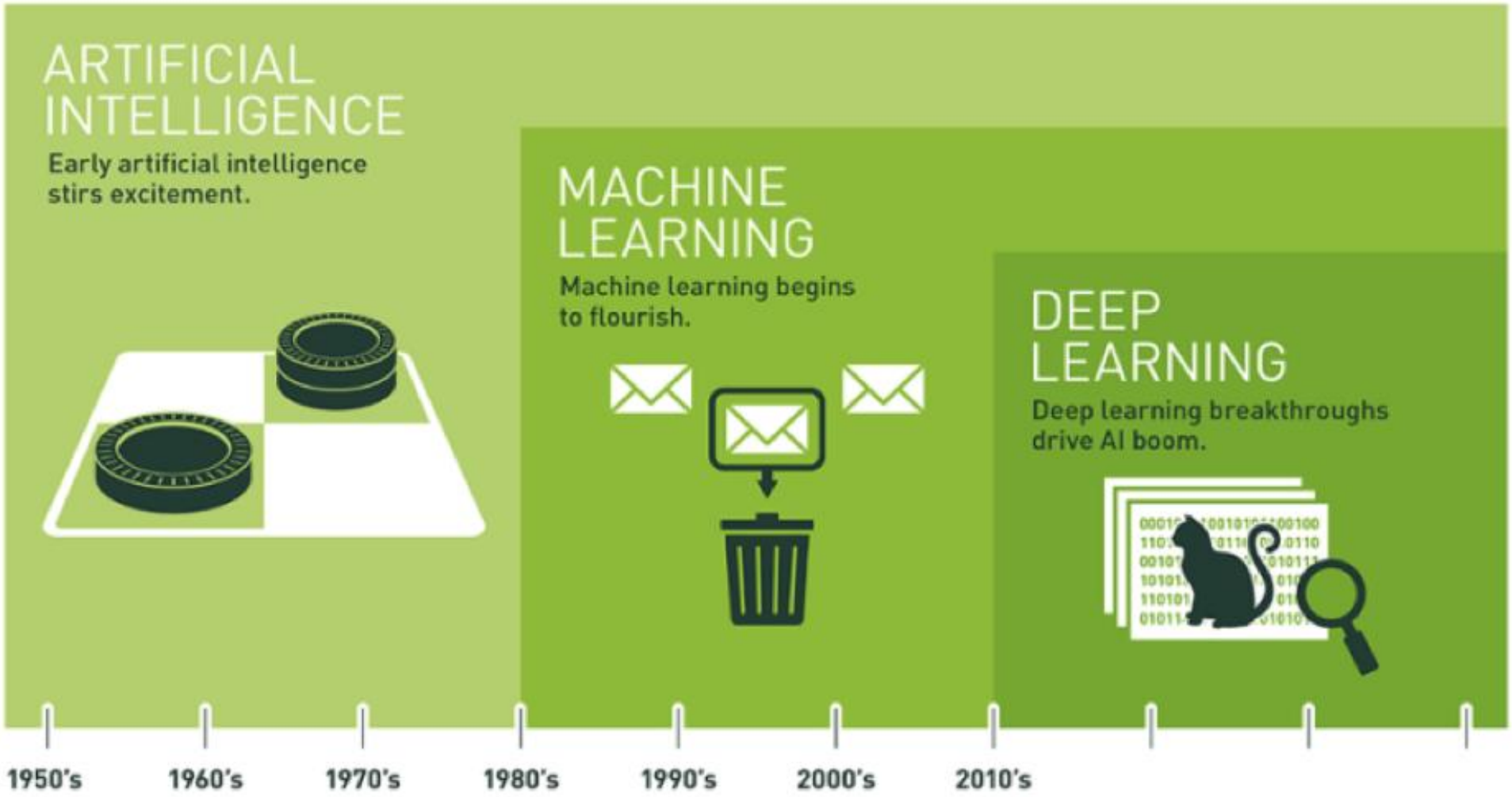
Buatlah sebuah Roleplay untuk 5G Trial/Field Test di Indonesia yang terdiri dari Regulator (KEMKOMINFO) dan Operator Telkomsel, Indosat, XL, Three dan Smartfren. Regulator (dengan masukan dari Operator) menentukan Operating Band pada **Frequency Ranges (FR)** FR1 & FR 2. Dan masing-masing operator mengajukan proposal 5G Trial/Field Test dengan mengisi tabel dibawah ini.

Operator Name :	FR 1	FR 2
Operating Band		
Channel Bandwidth		
SCS		
MIMO		

Operator Name :	enhanced Mobile Broadband (eMBB)	Ultra-Reliable Low Latency Communication (URLLC)	massive Machine Type Communicatio (mMTC)
Trial/Field Test Use Case			
Application			
Target KPI	Peak data rate 20 Gbps	Latency < 1 ms	1 million device per km^2
How to Test			

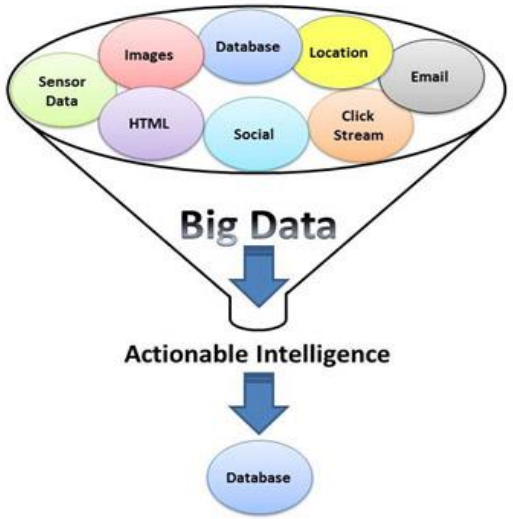
What The Other Technology

Artificial Intelligence / Machine Learning



Big Data

Big data is the term for a collection of data sets so large and complex that it becomes difficult to process using on-hand database management tools or traditional data processing applications. The challenges include capture, curation, storage, search, sharing, transfer, analysis, and visualization.



12+ TBs **of tweet data every day**





25+ TBs **of log data every day**



30 billion **RFID tags today (1.3B in 2005)**



76 million **smart meters in 2009… 200M by 2014**



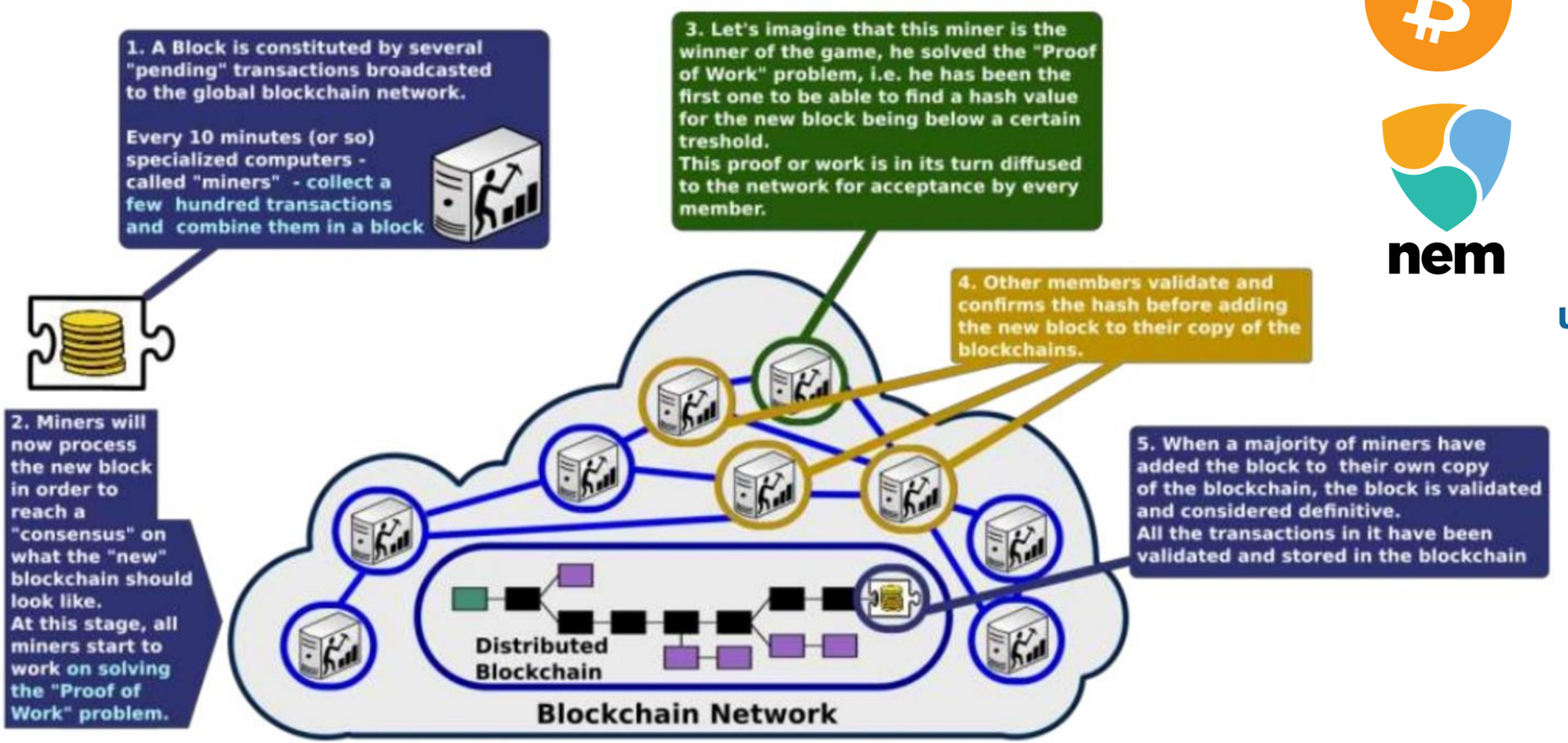
4.6 billion **camera phones world wide**

100s of millions of GPS enabled **devices sold annually**

2+ billion **people on the Web by end 2011**

2016?

Blockchain









wanchain

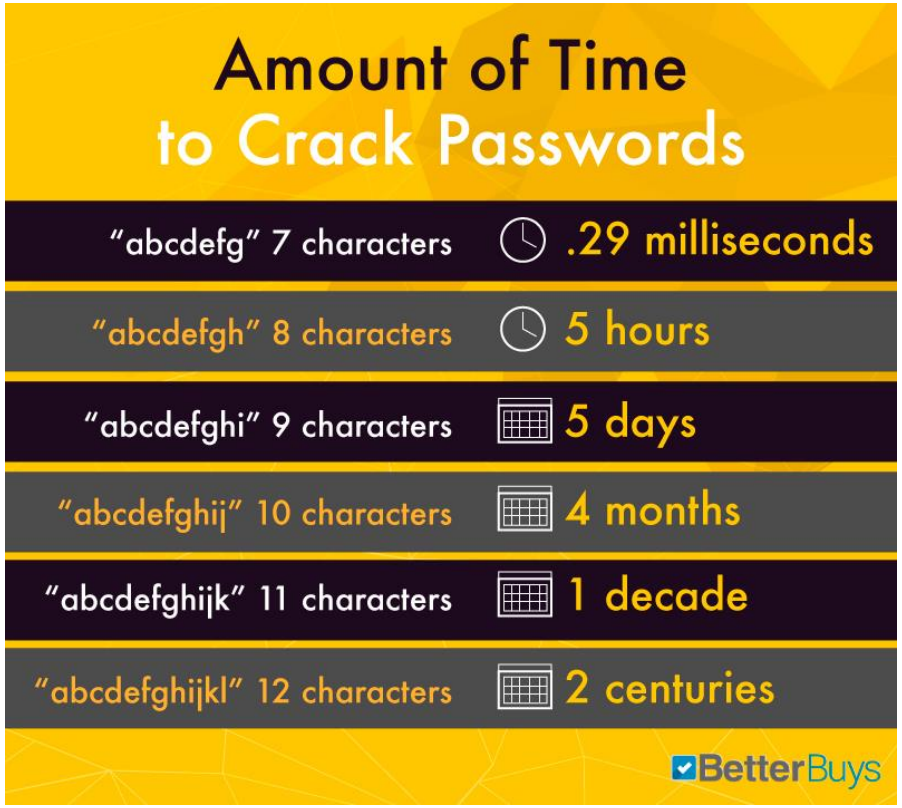
FLOATWAY SYSTEMS

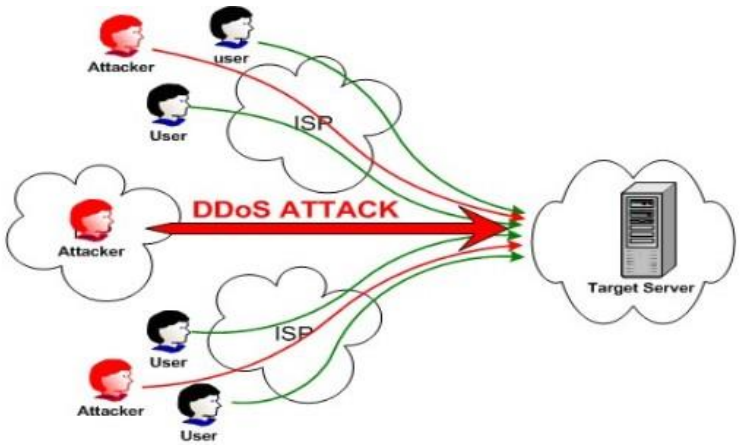
Cyber Security

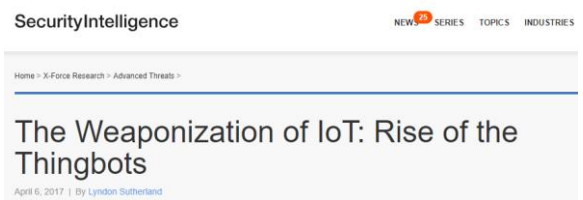












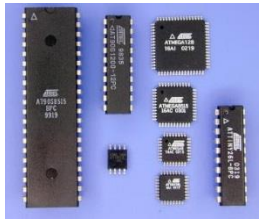


Internet of Things

An infrastructure of interconnected objects, people, systems and information resources together with intelligent services to allow them to process information of the physical and the virtual world and react.

IoT Device

1. Sensor
2. Actuator
3. Microcontroller
4. Firmware
5. Power





IoT Network

Short Range

1. Bluetooth
2. Zigbee
3. Wifi

Long Range

1. LoRa
2. Sigfox
3. Cellular Network

3GPP A GLOBAL INITIATIVE

LoRa

sigfox

IoT Platform

Cloud Computing

1. Infrastruktur as Service (IaaS)
2. Platform as Service (PaaS)
3. Software as Service (SaaS)

Azure

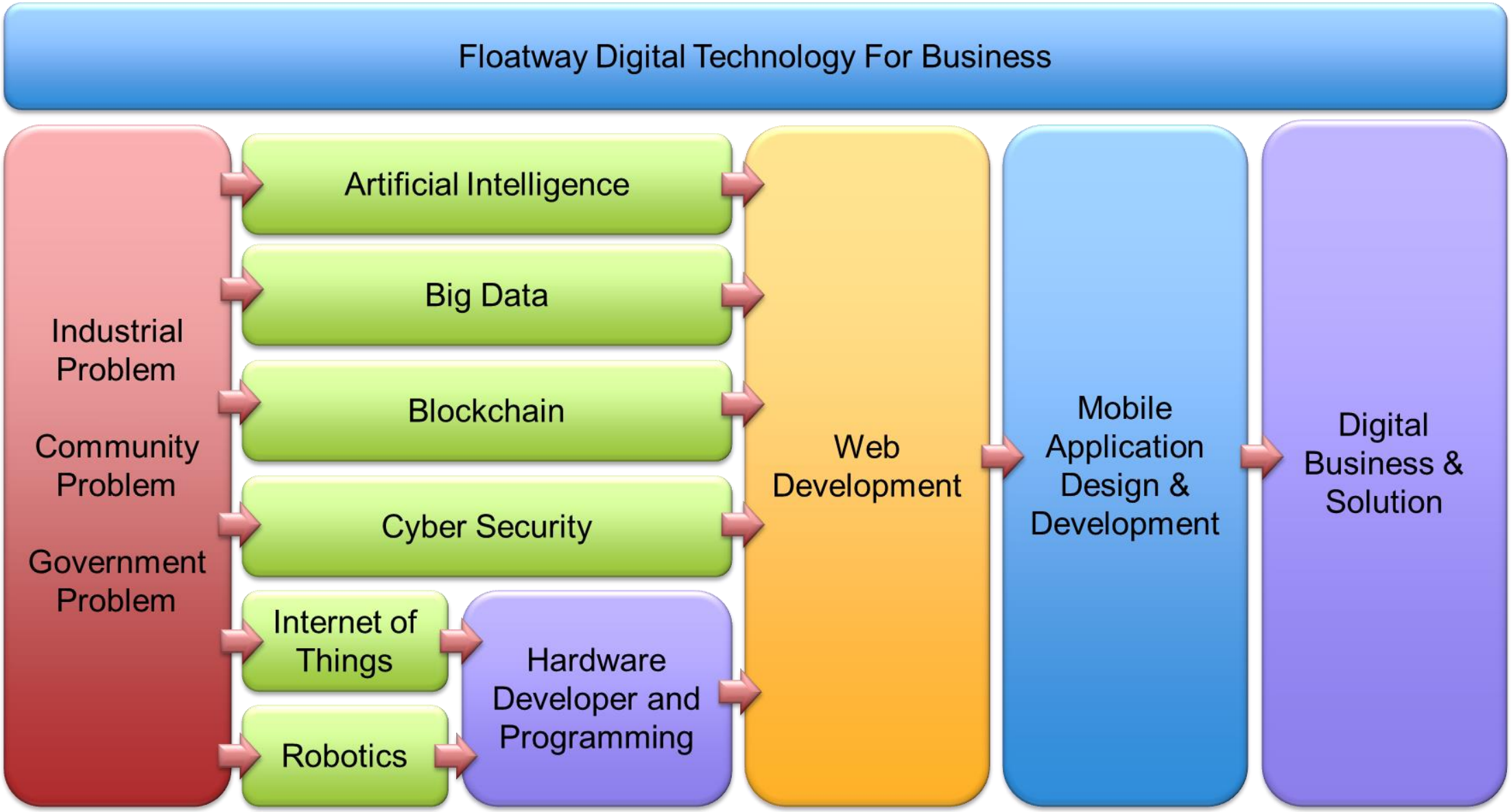
Google Cloud

aws

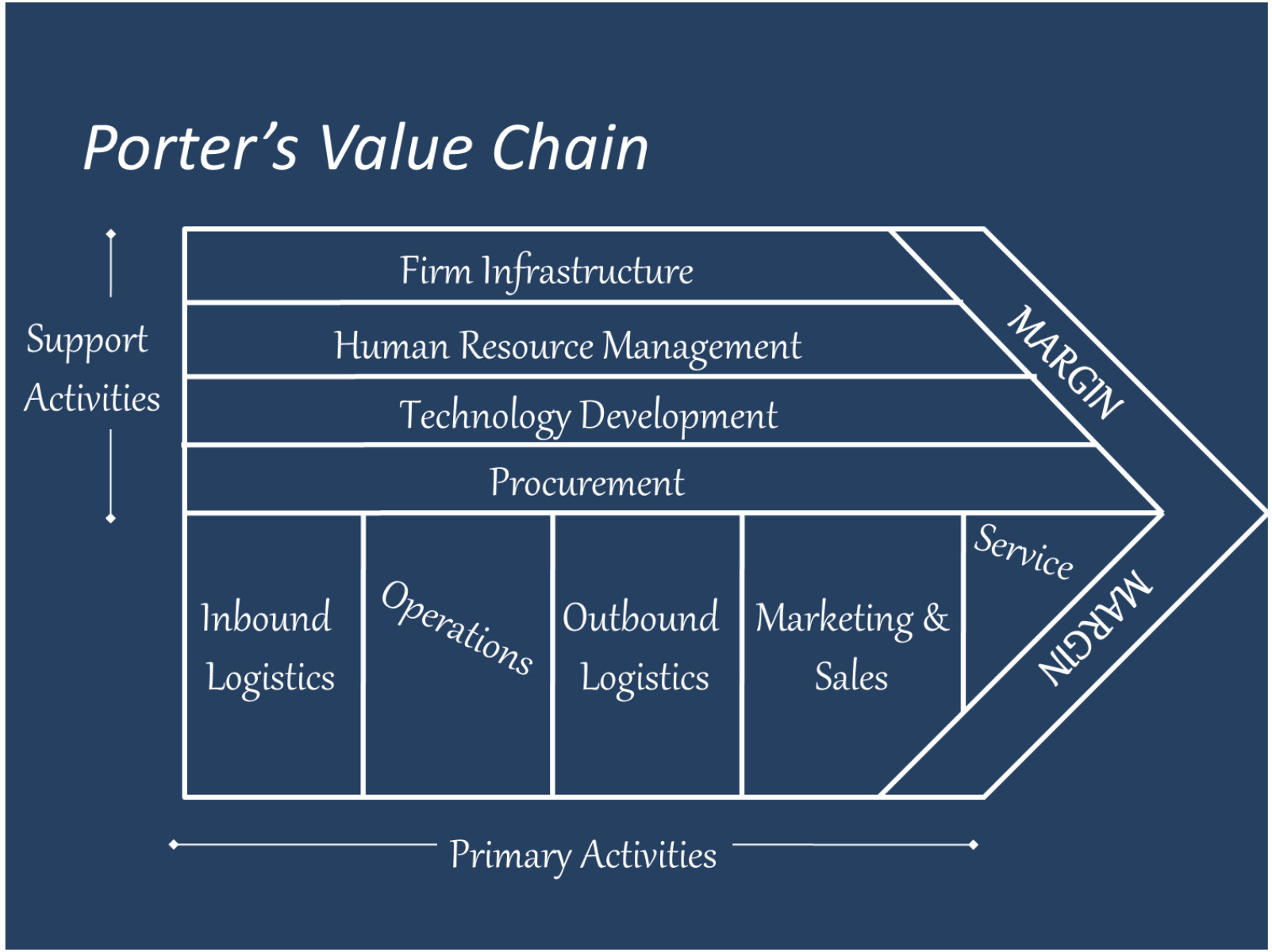
IoT Application

1. Smart Home
2. Smart Building
3. Smart City
4. Smart Car
5. Smart Agriculture/Farming
6. Smart Heathcare
7. Smart Grid
8. Industry 4.0

How Technology Can Improve Your Business


Floatway Digital Technology For Business
Industrial Problem
Community Problem
Government Problem
Artificial Intelligence
Big Data
Blockchain
Cyber Security
Internet of Things
Robotics
Hardware Developer and Programming
Web Development
Mobile Application Design & Development
Digital Business & Solution

Technology as Support



Technology as Core Business

Where to learn further about Technology



Machine Learning With Python

Training ini difokuskan untuk pemula yang ingin memiliki kemampuan untuk mempelajari Machine Learning secara mandiri menggunakan bahasa pemrograman Python.

Machine Learning merupakan cabang dari kecerdasan buatan, yang mencakup perancangan dan pengembangan algoritme yang memungkinkan komputer untuk mengembangkan perilaku yang didasarkan pada data empiris, seperti dari sensor data basis data. (wikipedia)

Syllabus

- Image Recognition
- Face Recognition
- Text Recognition
- Sentiment Analysis



Big Data Analytics

Syllabus

Introduction to Big Data Analysis
Installation / Anaconda, PyCharm, Gephi/
Introduction to Python Programming
Data Acquisition

- API
- Scrapping
- Headless Browser

Exploratory data Analysis

- Graphical Examination
- Simple Statistic
- Data Transformation

Machine Learning Concept (Training, Validation, Testing)
Theory, Hands-on :

- Data Forecasting
- Clustering
- Classification

Text Analytics

- Data Cleansing
- Segmentations
- Latent Semantic
- Network Analytics
- Preparation
- Exploring Gephi



Blockchain : Building Your Smart Contract & Ethereum Apps for Developers

Syllabus

Introduction
- What is Blockchain.
- What is Ethereum.
- Metamask Setup.
- Making Ethereum Accounts.
- Introduction to Solidity & Remix Editor.
- Ethereum Testing Network.

Building Simple Contract
- First simple contract.
- Gas & Transactions.
- Contract Deployment.
- Compiling contract.
- Installing NPM & Modules needed.
- Setup Web3 & Ganache Library
- Setup Infura Login.
- Setup Wallet Provider.
- Testing Contract.
- Verifying Etherscan.

Smart Contract
- Lottery Contract Overview.
- Basic Solidity Types.
- Overview Arrays & Structs
- Validation with require.
- Pseudo Random Generator.
- Selecting Winner.
- Sending Ether from Contract.
- Testing Contract with Mocha.

Real Live Ethereum App
- Ethereum App Architecture.
- Application Overview.
- Building Web Front End for Ethereum Contract.
- Testing Web Front End.

Internet of Things Business & Technology



Syllabus

1. WHAT is IoT
- IoT Definiton
- IoT Architecture
- IoT Standard (2G/3G/4G, LTE-M/eMTC, NB-IoT, Sigfox, LoRa)
- IoT Regulation in Indonesia

2. WHY is IoT
- Business Driven
- Technology Driven

3. WHO is IoT Player
- Indonesian Ecosystems

4. WHEN, WHERE and HOW To Create IoT Business Model

Practicing to create your IoT Business Model
- Defining Customer Segments
- Defining Customer Relationships
- Defining Channels
- Defining Value Propositions
- Defining Key Partner
- Defining Key Resources
- Defining Key Activities
- Defining Revenue Streams
- Defining Cost Structure

Practicing to create your IoT Timeline

Practicing to create your IoT Financial Projection

Simple and Awesome Home IoT System (SWIT)

1. What is SWIT?

2. Ingredients
- Hardware :
  - ESP32, Sensors (DHT22, Human Motion RCWL-0516, Fire Detection), Actuator (SSR)
- Software :
  - Atom, Arduino, Platformio, Python, NodeJS
- Connection :
  - WiFi, Bluetooth, AdafruitIO, IFTTT
- Where to get it?

3. Platformio
4. ESP32
5. DHT22
6. RCWL-0516
7. Fire Detection
8. SSR
9. AdafruitIO
10. IFTTT
11. SWIT Building
12. Finish



LET'S COLLABORATE

AND UPGRADE YOUR KNOWLEDGE WITH US!

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